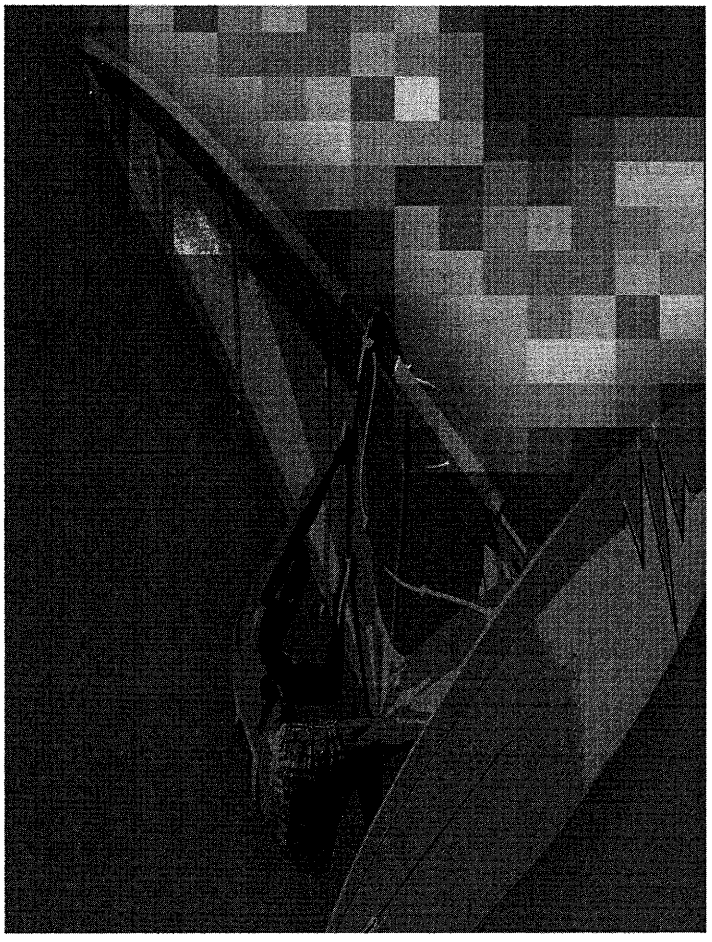
>>EXPERIMENTING with science FORCES & ELECTRICITY >>



Peter Gribben

Adapted from original material by David Palmer

INSTANT LESSONS

Peter Gribben is currently Head Teacher of Science and Agriculture at Crookwell High School in the Southern Tablelands of New South Wales. He has taught Science at all levels for around 30 years in England, the Middle East and Australia. Peter has written many science books; both conventional textbooks and photocopiable worksheets. He also has considerable experience in preparing examination material. Peter has a special interest in preparing science materials that are accessible to students of all levels.

Peter's personal interests include raising German Shepherds, restoring Minis and reading science fiction.

Acknowledgement

The author and publisher would like to thank Dr David Palmer for the use of his original material on which this series is based, and for his kind assistance in reviewing and contributing to this new series.

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Disclaimer

Every effort has been made to present activities in a safe way, giving warnings to teachers of potential hazards that can be foreseen. However, because requirements differ between states and often change over time, it is each teacher's responsibility to do their own risk assessment for any activity they choose to do with their classes.

The publisher and the author have made every reasonable effort to ensure that the experiments and activities in this book are safe when conducted as instructed, but assume no responsibility for any damage caused or sustained. Students should carry out all experiments under close supervision.

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INTRODUCTION

Science activities that have the element of surprise (that is, an unexpected result) are interesting to people of all ages. A surprising result can motivate students to examine their own ideas and to want to know 'Why is it so?'

Experimenting with Science is intended to stimulate student interest in science, but it is not about big bangs and explosions nor does it emphasise the use of scientific apparatus. In fact, a conscious decision was made to include many activities that do not rely on specialised equipment. When science involves everyday materials (and activities that can be done at home in the kitchen) then this automatically makes science relevant to real life.

Experimenting with Science is based on original material devised by Dr David Palmer of the University of Newcastle, New South Wales, who kindly reviewed and contributed to this new series.

The three books in this series are divided into broad topic areas: *The Physical World*, *The Living World* and *Forces and Electricity*.

The experiments are presented clearly with

- a brief introduction to the concept
- list of materials
- safety instructions
- clear, step-by-step procedure
- clear diagrams
- space for recording results
- questions for students to answer, included at key points to help them explain what they have seen and learnt, and also extend their understanding.

Answers are provided at the back of the book for easy checking and marking.

In all the experiments, there is little background knowledge assumed, which helps make them accessible to a wide range of ability levels. Observation and explanation skills are a major part of each activity, which contributes to the students' development of science skills. The short but related experiments in each of the topics make them ideal for small group tasks where students move between work stations, or for teachers to carry out the experiments in front of the class, involving the group as a whole to work through the tasks, discussing their observations as they go.

There are four icons that you will encounter through the book:



Teaching tips – where you see this icon, flip to the back of the book (page 48) for pointers about preparation, sourcing materials or comments on results.



These experiments use materials that may not be standard in your school laboratory, or require special preparation. For all the other experiments, basic materials only are needed.



These experiments are suited to independent work.



These experiments can easily be carried out by students at home.

There are detailed safety instructions for any experiments that might involve risk factors. Please note, however, that this book is designed for use by teachers in their classrooms, not for students in an unsupervised environment. Teachers should follow all normal safety procedures.

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1_MAGNETISM IN THREE DIMENSIONS

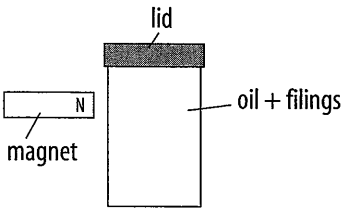
1.1 2-D or 3-D? T

Magnetic fields aren't flat and two-dimensional; they are three-dimensional.

Materials: paraffin oil, two bar magnets, iron filings, teaspoon, clear glass jar or plastic container with a close-fitting lid

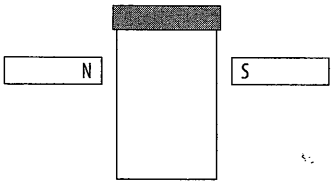
Procedure

- Pour paraffin oil into the container until it is almost full.
- Add a small amount of iron filings.
- Screw the lid on securely.
- Shake the container until the oil and filings are well mixed.
- Place one pole (North) of a single magnet next to the mixture as shown.



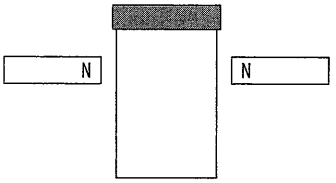
(A) *Draw the new arrangement of iron filings on the illustration.*

- Remove the magnet and shake the container until the filings are mixed again.
- Place unlike (North + South) poles next to the mixture as shown.



(B) *Now draw the new arrangement of iron filings on the illustration.*

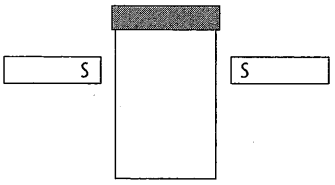
- Remove the magnets and shake the container until the filings are mixed again.
- Place like poles (North + North) next to the mixture as shown.



(C) *Now draw the new arrangement of iron filings on the illustration.*

(D) *Draw what you think would be the arrangement of the iron filings if you put two South poles next to the mixture as shown in the illustration.*

- Place like poles (South + South) next to the mixture as shown.
- Compare what you thought would happen with what actually happened.



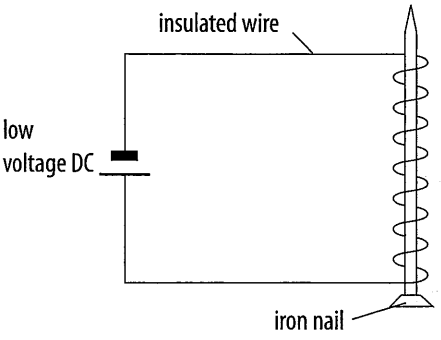
1.2 An electromagnet T

Electricity and magnetism are interconnected.

Materials: large iron nail, low voltage DC, insulated copper wire, steel paperclips

Procedure

- Wrap insulated wire around a nail.
- Connect the ends of the wire to low voltage DC.
- Place the end of the iron nail next to some paperclips.



(E) *Describe what happens to the paperclips.* _____

_____

- Switch off the current.

(F) *Describe what happens to the paperclips now.* _____

_____

(G) *What have you made?* _____

_____

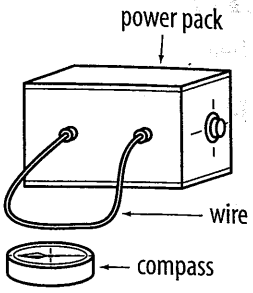
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2_INVISIBLE FIELDS

An electric current consists of moving electrons. When an electric current flows, it produces a magnetic field. In electromagnets magnetism can be switched on or off.

2.1 Fooling a compass T

A compass needle points to the strongest North Pole in its vicinity.



Materials: insulated wire, low-voltage DC power supply, compass

Procedure

- With the switch off, use an insulated wire to connect the two terminals of a power pack.
- Place a pocket compass next to the wire so that the wire and the compass needle are parallel.
- Set the power pack to 2 V and briefly switch on the current.

(A) *Describe what happens to the compass needle.* _____

(B) *Forecast what would happen if the direction of the current was reversed by switching the terminals on the power pack.*

_____

- Reverse the direction of the current.

(C) *Describe what happens to the compass needle.* _____

(D) *Forecast what would happen if the compass was placed on the other side of the wire.* _____

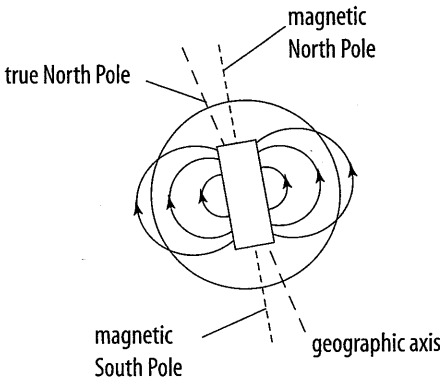
- Place the compass on the other side of the wire.
- Briefly switch on the current.

(E) *Describe what happens to the compass needle.* _____

(F) *Complete the passage to explain what you observed. Use these words: current, field, magnetic, off.*

A current flowing along a wire creates a _____ *field around it. The direction of the* _____ *determines the direction of the magnetic* _____. *The magnetic field disappears when the current is switched* _____.

The Earth's magnetic poles are moving at a rate of about 10 km each year. The southern magnetic pole is currently located on the coast of Antarctica, and the northern magnetic pole is located in northern Canada.



(G) *Outline the problems caused by not having the magnetic and geographical poles at the same location.* _____

_____

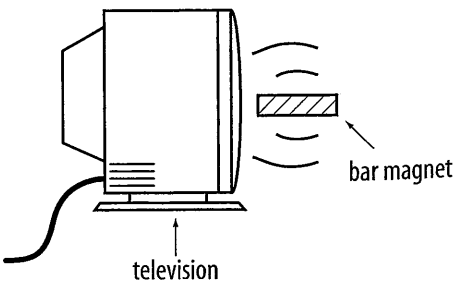
(H) *What is meant by the geographic axis of the Earth?* _____

_____

2.2 Changing the picture

The picture on a conventional (CRT) television is caused by electrons bombarding the screen.

Materials: television, bar magnet



Procedure

- Warning: try this on an old TV and do not hold the magnet in one place for too long as it could damage the TV tube.
- Switch on your TV and wait until you have a picture.
- Slowly move a strong bar magnet across the front of the screen.

(I) *Describe what you see.* _____

(J) *Explain what happened.* _____

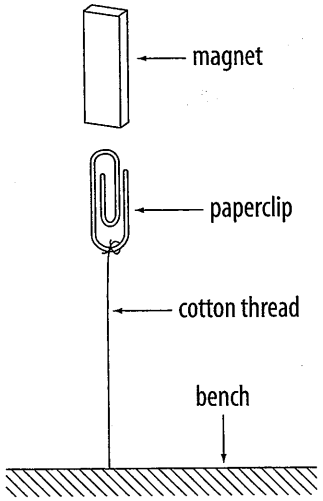
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3_ARE YOU ATTRACTED?

Forces move objects or change the shape of objects. Magnetism is a field force; it doesn't have to come into physical contact with an object to affect it. The only magnetic metal elements are iron (Fe), cobalt (Co) and nickel (Ni).

3.1 A paperclip in suspense

You can use a bar magnet and a paperclip to show the existence of a magnetic field.



Materials: light cotton thread, scissors, paperclip, bar magnet, sheets of paper, wood, plastic, aluminium foil, glass and steel

Procedure

- Cut about 30 cm of light cotton thread.
- Attach one end to the bench and tie the other end to a paperclip.
- Lift the paperclip up with a magnet until the cotton is taut.
- Move the magnet slightly away from the paperclip.

(A) *Describe what happens.* _____

- Move a sheet of paper through the space between the magnet and the paperclip.

(B) *Describe what happens.* _____

(C) *Complete the passage to explain the behaviour of the paperclip. Use these words: attracted, magnetic, pass, touch.*
The steel paperclip does not have to _____ *the magnet in order to be* _____ *to it – it just has to be within the* _____ *field. The magnetic field can* _____ *through non-magnetic materials, such as paper.*

Further investigation

Which other types of materials will the magnetic field pass through?

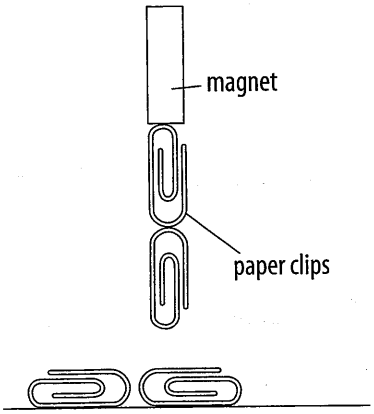
- Predict whether the paperclip will remain suspended if a particular material is placed between it and the magnet.
- Carry out the investigation.

(D) *Record your predictions and your actual results in the table.*

Material	Prediction	Test result
wood		
plastic		
aluminium		
glass		
steel		

3.2 How many paperclips?

Magnetism can be transmitted by steel.



Materials: magnet, steel paperclips

Procedure

- Place a paperclip on the end of a magnet.
- Touch another paperclip onto the first paperclip as shown.
- Keep adding a paperclip at a time until no more can be held by the magnet.

(E) *How many paperclips could be held by the magnet?* _____

(F) *Give the symbols for the following magnetic elements: iron* _____ *cobalt* _____ *nickel* _____

Some people have enough iron in their bodies to set off airport metal detectors. These people have a condition called haemochromatosis, a genetic disorder that results in iron overload.

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4_MORE MAGNETS

For over 2000 years people have used magnets to help them find their way around. A lodestone is a magnetic piece of iron used as a navigational aid. You can make your own lodestone.

4.1 Heading north

Materials: scissors, paper, cord or thread, bar magnet

Procedure

- Before you start this investigation, find out which way is north, and ensure no iron or steel objects are close by.
- Cut out a rectangle of paper big enough to act as a sling or cradle for a bar magnet.
- Place the magnet in the sling.
- Tie cord around the sling and suspend the magnet.
- Allow the magnet to swing freely.

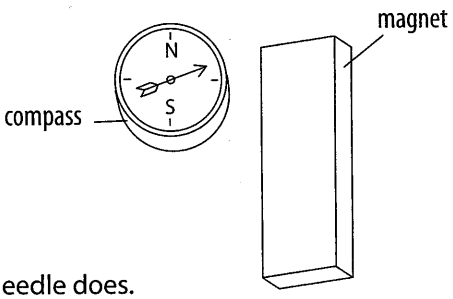
(A) *When the magnet had finished swinging, which way did it point?* _____

(B) *What is the bar magnet acting as?* _____

Many migratory animals, such as pigeons and salmon, have tiny magnetic particles in their brains, and they appear to use the Earth's magnetic field for navigation. Strangely, we humans have these particles too, but scientists don't know why.

4.2 Plotting lines of magnetism

Although you can't see lines of magnetic force, you can see how they act. The 'North' pole of a magnet is really a 'North-seeking' pole, as it points to the Earth's North magnetic pole.



Materials: bar magnet, small plotting compass

Procedure

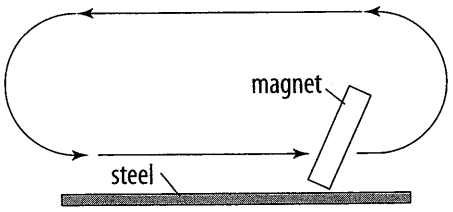
Note: ensure that the compass doesn't get too close to the magnet.

- Place the bar magnet on the bench away from any iron or steel objects.
- Place the compass near (but not touching) one end of the bar magnet.
- Move the compass around the bar magnet, observing what the compass needle does.

(C) *Describe what happens to the compass needle.* _____

4.3 Making a magnet

Materials: strong bar magnet, piece of steel (such as a hacksaw blade, nail or rod), steel paperclips



Procedure

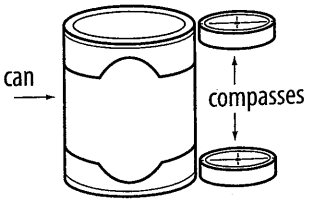
- Place your piece of steel on the desk.
- Pass the same pole over the piece of steel as shown, and repeat several times.
- Put your piece of steel near the paperclips.

(D) *What do the paperclips do?* _____

(E) *What have you made?* _____

4.4 Can it be magnetic? T

Materials: a steel food can that has been stationary for several weeks, compasses



Procedure

- Put the can on a table.
- Hold one compass next to the top of the can and another next to the bottom.

(F) *Describe what happens to the compass needles.* _____

(G) *Explain the behaviour of the compass needles* _____

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5_MAGNETISM ON THE MOVE

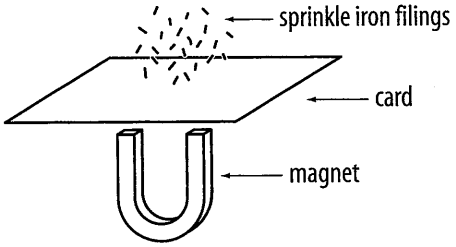
Iron filings can be used to help visualise the lines of force that surround a magnet.

5.1 Magnetic artwork T

Materials: horseshoe magnet, bar magnet, iron filings, card, aerosol can of varnish/hair spray

Procedure

- Position a horseshoe magnet under a sheet of light cardboard.
- Sprinkle iron filings onto the cardboard and tap it lightly.
- Spray the card with clear varnish or hair spray to preserve the pattern.
- Repeat the process using a bar magnet laid flat.



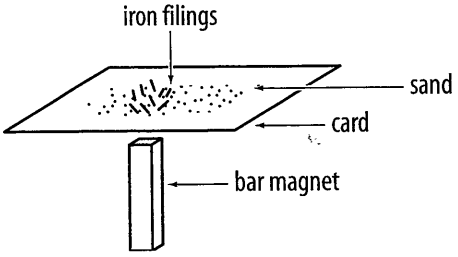
(A) *Describe how the two patterns created by the different magnets (horseshoe and bar magnets) differed.* _____

_____

5.2 Magnetic 'echidna'

You can make iron filings stand up and move around on a sheet of paper.

Materials: card, sand, iron filings, bar magnet



Procedure

- Mix equal quantities of sand and iron filings together.
- Place a teaspoonful of the sand and iron filings mixture on a sheet of card.
- Move a bar magnet under the card to separate the mixture into two heaps.

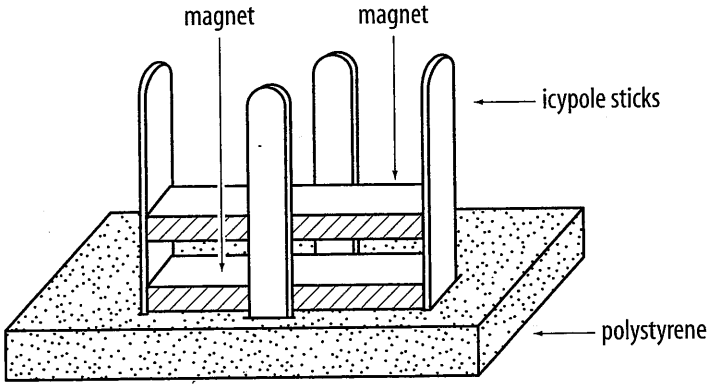
(B) *Describe what happens to the iron filings and sand mixture.* _____

(C) *Explain why the mixture behaved as it did.* _____

5.3 Levitate a magnet

You can show that magnets can repel as well as attract.

Materials: two bar magnets, polystyrene foam, four wooden icy pole sticks



Procedure

- Place a bar magnet on a thick slab of polystyrene foam.
- Insert an icy pole stick into the foam at each end of the magnet, and two more on each side as shown.
- Slip another magnet, similarly aligned, (i.e. North and South poles of both magnets are lined up) into the corral formed by the icy pole sticks.

(D) *Describe what happens to the upper magnet.* _____

(E) *Explain why the upper magnet behaves as it does.* _____

(F) *An electrician bought a new screwdriver that had a magnetic end. Which of the following would be attracted by the screwdriver: iron nail, brass screw, aluminium washer, copper wire, steel pin?* _____

(G) *A person tells you that they have a bar magnet with two South Poles. Would you believe them? Explain your answer.*

_____

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6_ELECTRICITY IN MOTION-A

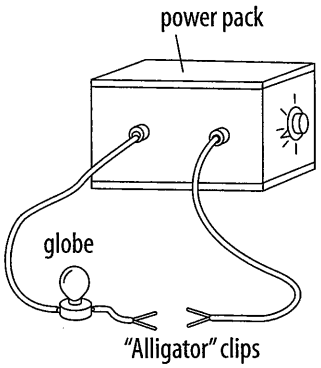
Electric current is the flow of electrons. Electrons carry a negative charge. In the laboratory we tend to use transformers ('power packs') or batteries, because mains electricity can be dangerous.

6.1 To boldly flow T

Some substances allow electricity to flow through them. These substances are called conductors. Materials which do not allow electricity to go through them are called insulators. This next experiment will help you to group substances into conductors or insulators.

Materials: power pack, globe, 3 insulated leads (2 with alligator clips), selection of solid materials

Procedure



- Set up the circuit as shown.
- Set the voltage to the lowest setting. (You may have to adjust this voltage, depending on the particular type of globe used.)
- Briefly touch the alligator clips to ensure that the globe is working.
- Attach one clip to one end of the material you are testing and the other clip to the other end.
- Note if the globe lights up.
- Remove the object and repeat the process with other materials.
- Note all your results and decide which materials are conductors and which are insulators.

(A) *Record your results in the table below.*

Name of material	Conductor or insulator

(B) *How did you decide whether a substance was a conductor or an insulator?* _____

(C) *What did the conductors all have in common?* _____

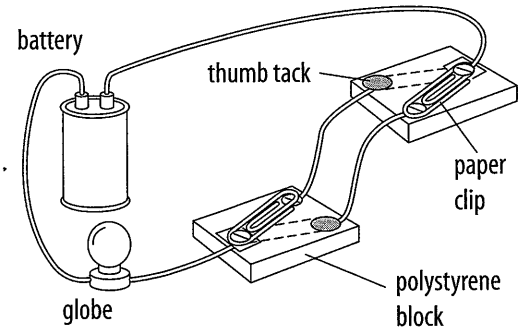
(D) *Complete the following passage. Use these words: glass, cables, insulators, protect.*

Copper is a very good conductor and is used in _____ *and house wiring. Materials which do not allow electricity to go through them are called* _____. *These include* _____, *plastics, rubber and wood. Plugs and sockets are made of insulators to* _____ *you.*

6.2 Making the switch

Switches turn electric current on and off. This experiment makes a special type of switch.

Materials: battery, drawing-pins, polystyrene (or wooden) blocks, globe, paperclips, connecting wires



Procedure

- Set up circuit as shown.

(E) *Does the globe light up?* _____

(F) *Explain your answer to (e).* _____

_____

- Move the paper clips one at a time to dotted positions shown on the polystyrene block.

(G) *What happens to the globe?* _____

(H) *Explain why the globe behaved as it did.* _____

(I) *Give a possible use for the switch you made.* _____

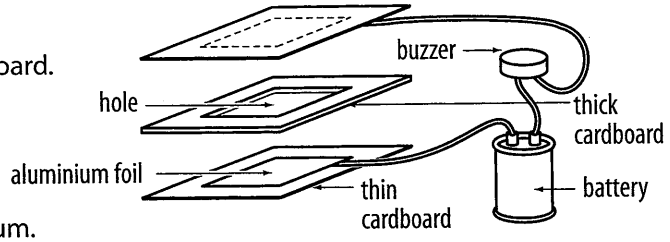
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6.3 Pressure switch

Some switches do not look like conventional switches.

Materials: 2 thin cardboard sheets, one thick cardboard sheet, scissors, aluminium foil, sticky tape, insulated wire, buzzer, battery

Procedure



- Tape aluminium foil to one side of a piece of thin cardboard.
- Tape the exposed end of a wire to the aluminium.
- Repeat this process to make another one.
- Cut a hole in a thick piece of cardboard and place it between the two pieces of thin cardboard and aluminium.
- Use wires to connect the pressure switch to a buzzer and a battery.
- Place the pressure switch flat on the bench and press the top piece of cardboard.

(J) *What happens?* _____

(K) *Explain your result.* _____

_____

(L) *Give a possible use for this kind of pressure switch.* _____

6.4 Bi-metal strip

A bi-metal strip consists of two different metals joined together. Metals expand when heated.

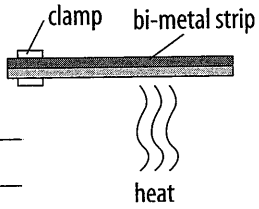
Materials: Bunsen burner, bi-metal strip, retort stand and clamp

! SAFETY NOTE !

The bi-metal strip will get very warm. Allow it to cool before touching it.

Procedure

- Hold a bi-metal strip steady using a retort stand and a clamp.
- Heat the bi-metal strip with a Bunsen flame as shown for two minutes.
- Allow it to cool.



(M) *Describe what happens to the bi-metal strip when it is being heated.* _____

_____

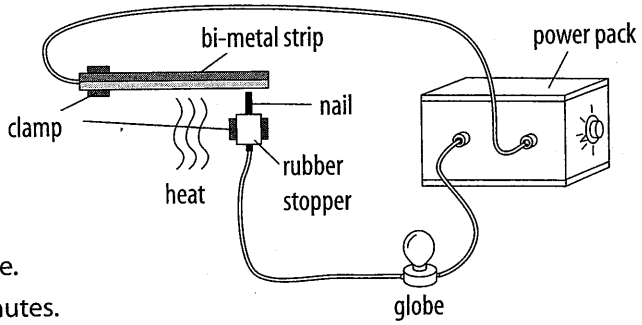
(N) *Describe what happens to the bi-metal strip when it cools down.* _____

(O) *Explain why the bi-metal strip behaves as it does when heated.* _____

6.5 Bi-metal switch T

Materials: Bunsen burner, bi-metal strip, retort stands and clamps, power pack, globe, insulated wires, nail, rubber stopper

Procedure



- Place a steel nail through the hole in a rubber stopper.
- Set up the apparatus as shown.
- Ensure that the bi-metal strip is set up so that it will bend downwards when heated.
- Set the power pack at an appropriate voltage for the globe.
- Using a Bunsen burner, heat the bi-metal strip for two minutes.

(P) *Describe what happens to the globe.* _____

(Q) *Explain your results.* _____

(R) *Give a possible use for the apparatus shown.* _____

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7_CIRCUITS WITH A DIFFERENCE

In a circuit, the electrons move around it. This flow of current can be used to do many things.

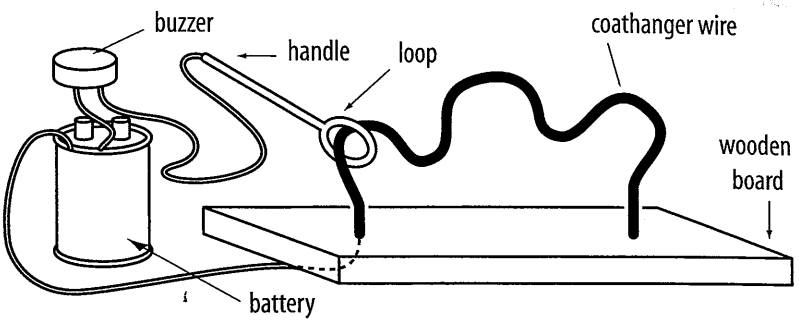
7.1 Steady as she goes

You can test your hand–eye coordination in this experiment.

Materials: wooden board, coat hanger wire, insulated wire, solid copper wire, battery, buzzer

Procedure

- Make a handle and loop out of solid copper wire.
- Set up the circuit as shown.
- Check that the buzzer will sound when the loop touches the bare wire.
- Discuss how this device could be used to compare hand–eye coordination.



(A) *Outline how you aim to carry out the experiment.* _____

_____

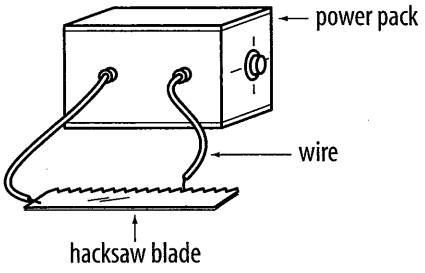
(B) *Record your results.* _____

_____

7.2 I saw it

An intermittent electric current behaves in a different way to a steady current.

Materials: power pack, insulated wire, hacksaw blade



Procedure

- Strip the ends of two pieces of insulated wire.
- Attach one end of each piece of insulated wire to the terminals of a power pack set on 4 V. (Make sure it is switched off.)
- Attach the other end of one wire to the end of a hacksaw blade.
- Darken the room and slowly run the bare end of the other wire along the serrated edge of the hacksaw blade.

(C) *Describe what happens.* _____

(D) *Complete the following passage to explain what happened. Use these words: steel, sparks, current.*

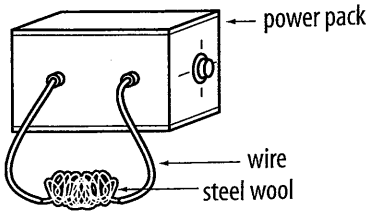
The intermittent flow of _____ *produces a series of small electric sparks, which strike the steel, causing some of the* _____ *particles to become white hot and break free from the blade as a visible spray of* _____.

7.3 Electrifying steel wool

Steel 'wool' is made up of fine strands of steel wire. It is used for cleaning.

! SAFETY NOTE !
Do this experiment on a heat-proof surface.

Materials: power pack, steel wool



Procedure

- Take a piece of steel wool about 5 cm long.
- Spread out the steel wool.
- Connect the steel wool to the power pack. (Make sure it is switched off.)
- Set the power pack to 10 V and switch it on for a few seconds.

(E) *Describe what happens.* _____

(F) *Explain why the steel wool behaves as it does.* _____

_____

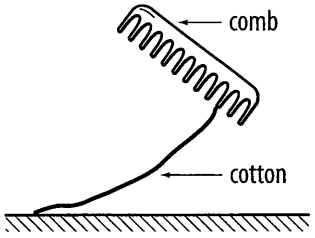
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8_GETTING CHARGED (ELECTROSTATICS)-A

Electric current is the flow of electrons. Static ('non-moving') electricity is caused when there is an imbalance of electrical charge. The two types of charge are positive (+) and negative (–). An electrical charge can sometimes be induced in an object by rubbing it. Electrical charge can cause objects to be attracted or repelled. Remember: 'opposites attract, like repels like'. Static electricity experiments work best on dry days. On damp days, some of the charge tends to leak into the humidity in the air.

8.1 Static snake

Materials: scissors, cotton thread, sticky tape, plastic comb, cloth



Procedure

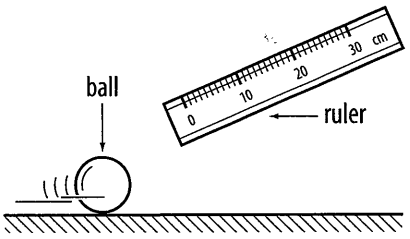
- Cut about 40 cm of cotton thread.
- Tape one end of the cotton to the desk.
- Rub a plastic comb with a cloth.
- Bring the rubbed comb close to the cotton.

(A) *Describe what happens.* _____

(B) *Explain your result.* _____

8.2 Follow me

Materials: plastic ruler, woollen cloth, table-tennis ball



Procedure

- Place a table-tennis ball on a level surface.
- Rub a plastic ruler with a woollen cloth.
- Bring the rubbed part of the ruler close to the table-tennis ball.

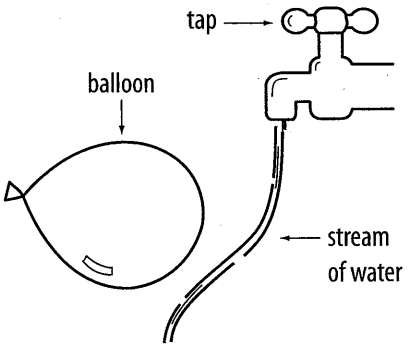
(C) *Describe what happens.* _____

(D) *Would you get the same result if you used a tennis ball? Explain your answer.* _____

_____

8.3 Bendy water

Materials: tap, balloon, cloth



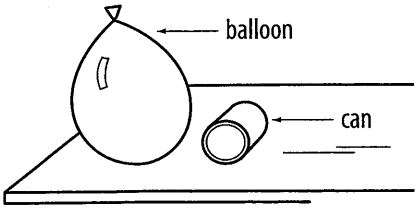
Procedure

- Blow up a balloon and tie the neck.
- Turn a tap on to produce a thin, steady, unbroken stream of water.
- Rub the balloon with the cloth.
- Bring the rubbed balloon close to the stream of water.

(E) *Describe what happens to the water.* _____

8.4 Can a balloon be attractive?

Materials: balloon, empty aluminium soft drink can, paper towel



Procedure

- Blow up a balloon and tie the neck.
- Place an empty aluminium soft drink can on its side on a level surface.
- Rub the balloon with the paper towel.
- Bring the balloon close to the aluminium can and then move it away. (Don't let the can and balloon touch.)

(F) *Describe what happens to the aluminium can.* _____

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8_GETTING CHARGED (ELECTROSTATICS)-B

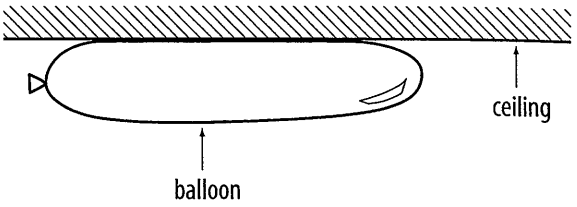
Are the following statements true or false?

(G) *A positive charge will repel a negative charge.* _____

(H) *Experiments with static electricity work best on dry days.* _____

8.5 Look, no hands!

Materials: sausage-shaped balloon, paper towel



Procedure

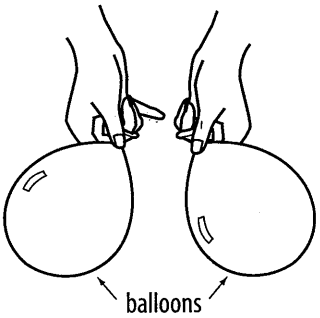
- Blow up a sausage-shaped balloon and tie the neck.
- Rub one side of it vigorously with paper towel.
- Hold the balloon so the rubbed part is touching the ceiling.
- Let the balloon go.

(I) *What happens to the balloon?* _____

(J) *Give an explanation for the balloon's behaviour.* _____

8.6 Repulsive balloons

Materials: two balloons, paper towel



Procedure

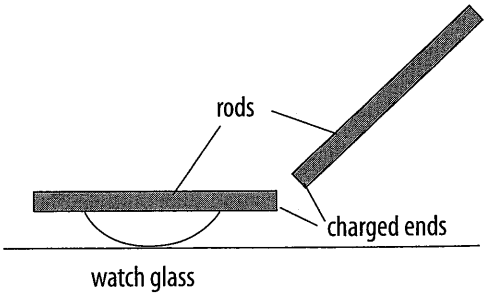
- Blow up two balloons and tie their necks.
- Rub both balloons vigorously with paper towelling.
- Hold the balloons by the necks, so they can hang and swing freely.
- Bring them near to each other.

(K) *Describe what happens.* _____

(L) *Explain why the balloons behaved in this way.* _____

8.7 Rotating rods

Materials: perspex rods, ebonite rods, woollen cloth, large watch glass



Procedure

- Place a watch glass on a level surface.
- Rub one end of an ebonite rod with a woollen cloth.
- Carefully balance this rod on the watch glass.
- Rub one end of another ebonite rod with a woollen cloth.
- Bring the two rubbed ends together as shown.

(M) *Describe what happens to the rods.* _____

- Repeat the procedure using different combinations of rods, charged with different types of cloth.
- Record your results in the table below.

Rod on watch glass	Rubbed with	Rod brought near watch glass	Rubbed with	Action of rods

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9_THE VAN DE GRAAF GENERATOR

A Van de Graaff generator is a machine for the generation of extremely high voltages. It was developed in the 1930s by American physicist Robert Van de Graaff. It consists of a high-voltage terminal, often an aluminium sphere, mounted at the top of an insulating column. A belt runs from a pulley at the base of the column to a pulley inside the sphere. A metal comb is mounted parallel to the moving belt and it puts electrons onto the belt. The belt carries its charge up into the sphere, and passes it to the sphere.

Ⓐ *Why must the sphere be placed on an insulating column?* _____

_____

! SAFETY NOTE !

Teachers should be careful when using a Van de Graaff generator. Instructions for the individual machine should be read carefully before using. An insulated earthing rod should be used to discharge the dome after each demonstration.

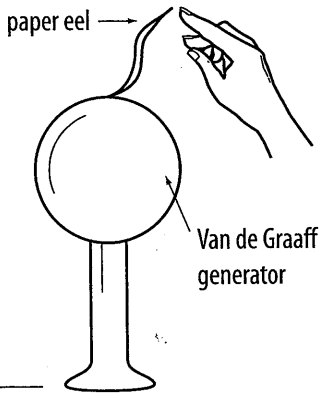
9.1 Electric eel (Teacher demonstration only)

You can make a paper 'eel'.

Materials: Van de Graaff generator, paper, scissors, sticky tape

! SAFETY NOTE !

Do not place fingers too close to the Van de Graaff generator or you may receive a shock.



Procedure

- Cut an eel shape about 30 cm long and 1 cm wide from thin writing paper.
- Use sticky tape to attach its tail to the dome of a Van de Graaff generator.
- Switch the generator on (at low speed only, if possible).

Ⓑ *What does the 'eel' do?* _____

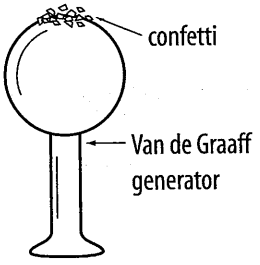
- Carefully extend your hand towards the electric eel.

Ⓒ *What does the 'eel' do now?* _____

Ⓓ *Explain the behaviour of the 'eel'.* _____

9.2 Flying confetti

Confetti is made up of small pieces of coloured paper. It is traditionally thrown during weddings.



Materials: Van de Graaff generator, confetti (or pieces of paper from a paper punch)

Procedure

- Place a handful of confetti on top of the dome of a Van de Graaff generator.
- Switch the generator on.

Ⓔ *Describe what happens to the confetti.* _____

Ⓕ *Explain the behaviour of the confetti.* _____

9.3 Flying saucers

Aluminium pie dishes can be put to a novel use.

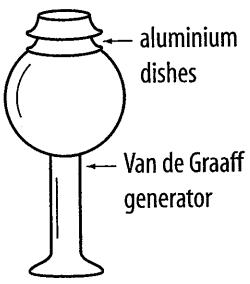
Materials: Van de Graaff generator, six aluminium pie dishes

! SAFETY NOTE !

Do not touch the generator or pie dishes while the generator is on.

Procedure

- Place an aluminium dish upside down on the dome of the Van de Graaff generator.
- Carefully place another five dishes on top of it.
- Turn the generator on.



Ⓖ *Describe what happens to the pie dishes.* _____

Ⓗ *Explain the behaviour of the pie dishes.* _____

_____

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10_ORGANIC ELECTRICITY

Electrical cells (sometimes called batteries) have two electrodes (made of different conductors) and an electrolyte to allow electrons to flow. An electrolyte is a liquid containing ions (charged particles). In a cell made of two metal electrodes, the more reactive metal is the negative (cathode) and the less reactive metal is the positive (anode). Did you know you can generate an electrical current with fruit?

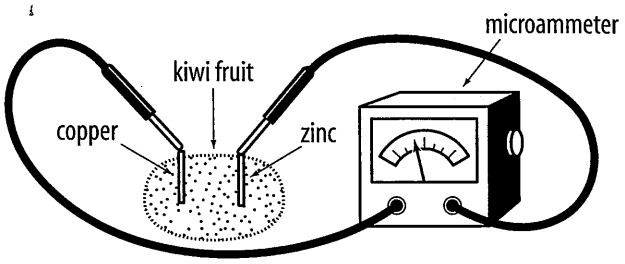
10.1 Electricity from a kiwi fruit T

You can use a kiwi fruit or a lemon to make an electric cell and also to work out the relative reactivities of metals. A microammeter will indicate the flow of current.

Materials: kiwi fruit (or lemon), metal electrodes (zinc, copper, iron/steel, nickel, lead), microammeter, electrical leads

Procedure

- Use an electrical lead to connect a copper plate to one terminal of a microammeter.
- Connect a zinc plate to the other terminal.
- Cut two slits about 2 cm apart in a kiwi fruit.
- Insert the two metal plates into the slits.
- Record which electrode is positive and which is negative.
- Note the reading on the microammeter.
- Repeat with other combinations of electrodes.



(A) *Record your results in the table below.*

Anode (+)	Cathode (–)	Meter reading

The most reactive metal will be the one that is the cathode the highest number of times.

(B) *Which is the most reactive metal?* _____

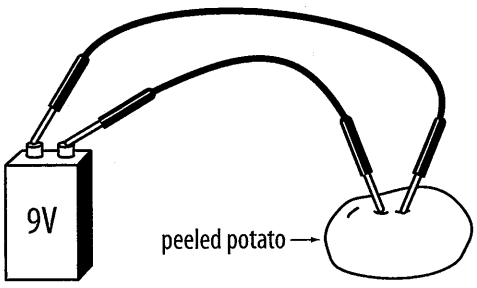
(C) *Which is the least reactive metal?* _____

(D) *Outline how you ensured that this was a fair comparison of the electrodes.* _____

_____

10.2 Potato battery tester

A potato can be used to test if there is some charge left in a battery and also which terminal is positive and which is negative.



Materials: peeled potato, 9 V battery, two pieces of insulated copper wire

Procedure

- Attach two wires to the terminals of a 9 V household battery.
- Insert the other ends of the wires into a peeled potato about 2–3 mm apart.

(E) *What happens at the negative terminal?* _____

(F) *What happens at the positive terminal?* _____

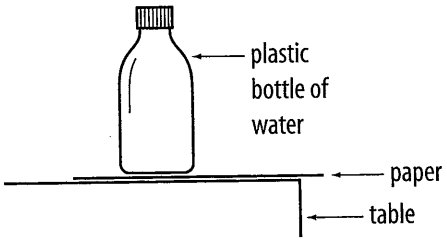
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11_I DON'T WANT TO CHANGE

Newton's first law helps to define inertia. An object will stay in one place (at rest) or in uniform motion in the same straight line unless acted upon by some external force. Inertia is a measure of an object's resistance to change. The greater an object's mass, the greater the inertia it will have.

11.1 Tablecloth trick

You can pull a sheet of paper out from under a bottle of water.



Materials: table, sheet of paper, plastic bottle of water

Procedure

- Stand a plastic bottle, full of water, on top of a sheet of paper, as shown.
- Slowly pull the paper.

(A) *Describe what happens.* _____

- Return the bottle and paper to their previous positions.
- Quickly jerk the paper out from under the bottle.

(B) *Describe what happens.* _____

(C) *Explain why the bottle behaved as it did.* _____

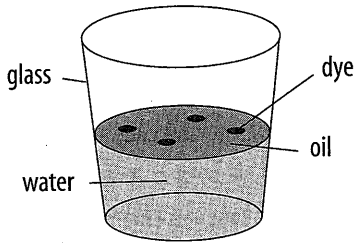
_____

(D) *What would happen if you used an empty plastic bottle? Explain your answer.* _____

_____

11.2 Inertia in action

Not only solid objects can have inertia. Liquids can too!



Materials: glass (or beaker), water, cooking oil, food dye, dropper

Procedure

- Half fill a glass (or beaker) with water.
- Pour a thin layer of cooking oil on the surface of the water.
- Place a few drops of food dye on the surface of the oil.
- Hold the top of the glass and rotate it slowly.

(E) *What happens to the contents of the glass?* _____

- Now rotate the glass quickly.

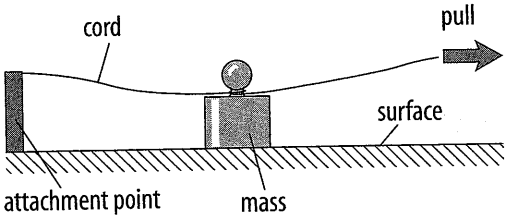
(F) *Now what happens to the contents of the glass?* _____

11.3 Breaking point

Where a string breaks can depend on how it is pulled.

Materials: cord (should be able to snap relatively easily), attachment point (such as a tap), a 500g mass

Procedure



- Tie the cord to the attachment point and the mass, leaving one end to pull, as shown.
- Pull the cord slowly and note where it breaks.
- Replace the cord.
- Pull the cord quickly and note where it breaks.

(G) *What happens to the cord?* _____

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12_BUMPING INTO EACH OTHER-A

An object that is moving possesses momentum. This is a product of mass times velocity. Newton's second law of motion states that the rate of change of momentum of an object is proportional to the force acting on the object. When objects collide elastically, the total momentum remains constant.

12.1 Basketball and tennis ball

This experiment shows what happens when two objects travelling at similar speeds but with different masses interact.

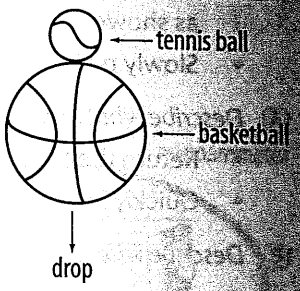
Materials: basketball, tennis ball, hard surface

! SAFETY NOTE !

This activity should be done outside in a clear area with a hard, flat surface.

Procedure

- Hold a tennis ball on top of a basketball at about chest height.
- Drop the two balls, still in contact with each other, onto a hard, flat surface.



(A) *Describe what happens.* _____

_____

(B) *Explain the behaviour of the balls.* _____

_____

Earth regularly collides with objects from outer space. About 500 meteorites hit the ground every year. A meteorite is an object from space that strikes the Earth's surface. The largest known meteorite weighs 61 tonnes.

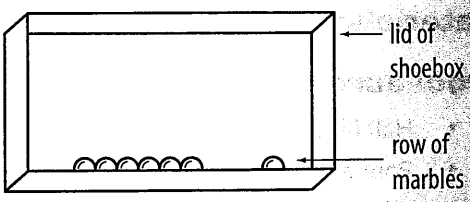
(C) *Would any of the meteorites mentioned affect the Earth's motion? Explain your answer.* _____

_____

12.2 Losing your marbles

Marbles are small spheres made of glass. They are very useful to demonstrate elastic collisions.

Materials: 8 marbles, shoe box



Procedure

- Put six marbles in the lid of a shoebox.
- Tilt it so the marbles are in a row along one edge.
- Make sure the marbles are all touching.
- Roll another marble along the edge so it collides with the end of the row.

(D) *Describe what happens.* _____

- Roll two marbles together to collide with the end of the row.

(E) *Describe what happens.* _____

Creative literacy

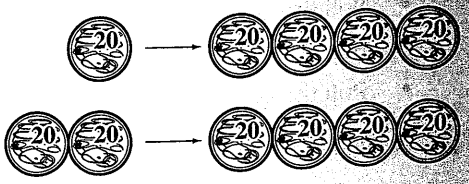
(F) *Many cartoon characters are involved in amazing collisions. Describe one example you believe to be scientifically inaccurate. Explain your reasons.* _____

_____

12.3 Colliding coins

Coins can be used to illustrate elastic collisions.

Materials: six 20-cent coins, smooth surface



Procedure

- Place four 20-cent coins in a row.
- Make sure the coins are all touching.
- Flick another coin so that it collides with the end of the row.

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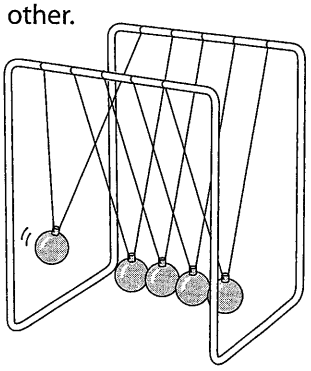
(G) *Describe what happens.* _____

- Repeat the procedure, but use two coins.

(H) *Describe what happens.* _____

12.4 Newton's cradle

Newton's cradle consists of a row of steel balls suspended by wires or cords and touching each other.



Materials: Newton's cradle

Procedure

- Pull back one ball and let it go.
- Allow the system to settle down.

(I) *Describe what happened.* _____

_____

- Pull back two balls and let them go.
- Allow the system to settle down.

(J) *Describe what happened.* _____

12.5 Bouncing back

Do all balls bounce as well as each other? The more elastic a collision is, the more momentum is conserved. In an elastic collision, the materials deform at the moment of collision and then return to their original shape.

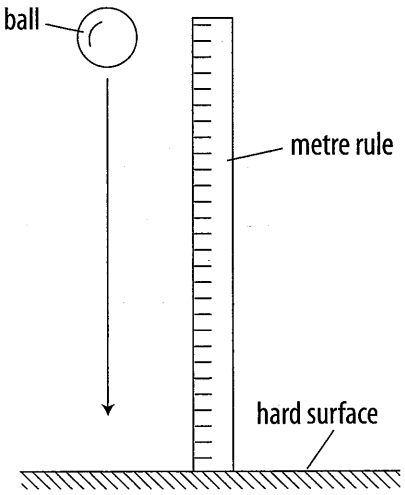
! SAFETY NOTE !

This activity should be done outside in a clear area with a hard, flat surface.

Materials: hard surface, tennis ball, golf ball, table-tennis ball, ball made of plasticine, metre rule

Procedure

- Hold a ball one metre above the ground.
- Drop the ball and note how far it initially bounces back.
- Repeat twice with the same ball and work out the average rebound distance.
- Repeat with all the other balls.
- Record your results in the table below.



Type of ball	Average initial bounce-back distance (cm)

(K) *No ball will ever bounce completely back to the original dropping point. Explain why this is the case.* _____

_____

(L) *Which collision is the most elastic?* _____

(M) *Which collision is the least elastic?* _____

(N) *Explain why the material mentioned in (m) was the least elastic.* _____

Research

(O) *Find out why modern cars have 'crumple zones'.* _____

_____

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13_ROCKET SCIENCE-A

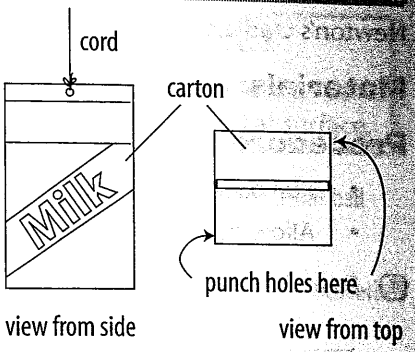
Sir Isaac Newton's third law tells us that for every action there is an equal and opposite reaction. In other words, forces work in pairs.

13.1 Water power T

> **! SAFETY NOTE !**
> This experiment should be done in a sink.

A milk carton can be used to illustrate Newton's third law in a very simple way.

Materials: empty cardboard milk carton, cord, pencil, water, sink, nail



Procedure

- Open up the top and rinse out a cardboard milk carton.
- Fill it with water.
- Reseal the top of the carton.
- Use a nail to punch a hole in the top part of the carton and attach a cord.
- Suspend the carton inside the sink.
- Using a pencil, punch two diagonally opposite holes in the sides of the carton, near the base, as shown.

(A) *Describe what happens.* _____

(B) *Explain, with reference to Newton's Third Law, why the carton behaved in the way that it did.* _____

_____

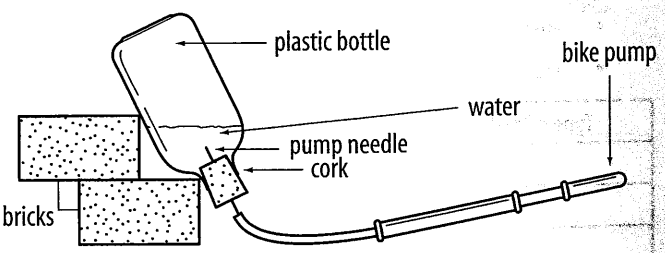
(C) *If you suddenly turn the water on full force on a garden hose and the far end is not secured, then the nozzle springs back and moves around on the ground. Explain why this happens.* _____

13.2 Get ready for lift off (Teacher demonstration only)

Rockets rely on a force coming out of the vehicle that causes it to be pushed in the opposite direction. You can use a bicycle pump to launch a plastic bottle.

> **! SAFETY NOTE !**
> - Teachers – wear safety glasses.
> - Do this activity outside in an open area.
> - The 'rocket' should be behind a safety screen.
> - No one should be in front of the bottle.

Materials: two bricks, plastic bottle, bicycle pump, water, needle inflator (as used in basketballs), safety glasses, safety screen



Procedure

- Pour water into a plastic bottle until it is about 1/8 full.
- Insert a pump needle through a cork and fit it into the neck of the bottle.
- Invert the bottle and attach a bicycle pump to the needle.
- Pump air into the bottle until it launches.

(D) *Describe what happens.* _____

- Repeat the experiment with various amounts of water.

(E) *Which amount of water gave the 'best' result?* _____

(F) *Explain why the rocket works.* _____

_____

When rockets and satellites come back from space, they often do so suspended by a parachute.

(G) *Why are parachutes used for these vehicles?* _____

(H) *Design and develop a way to attach a plastic-bag parachute to the bottle rocket so that it will open as the bottle begins to descend. List materials used, explain how it would work and draw a labelled diagram of your apparatus in your workbook.*

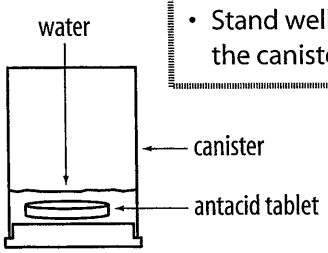
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13.3 Canister rocket (Teacher demonstration only)

You can make a film canister blast off. Antacid tablets generate carbon dioxide when they come into contact with water.

! SAFETY NOTE !
- Teacher should wear safety glasses.
- Stand well clear of the canister.

Materials: plastic 35 mm film canister (one where the cap fits inside, rather than over the outside; the cap should 'snap' on), antacid tablet, water



Procedure

- Add water until the canister is about 1/4 full.
- Add an antacid tablet then quickly put the lid on.
- Place the canister upside down on the floor and stand clear.

(I) *Describe what happens.* _____

(J) *Explain why the canister behaves as it does.* _____

_____

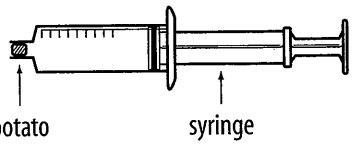
(K) *Animals developed rocket propulsion long before humans. Squid and octopus are propelled by squirting water from a funnel near their tentacles. Their ancestors were doing this long before humans appeared on the planet. How does this method of propulsion differ from the way most other fish move?* _____

_____

13.4 Propelled potato (Teacher demonstration only)

Air pressure can be used to shoot a piece of potato.

Materials: plastic syringe, potato



! SAFETY NOTE !
Wear safety glasses.

Procedure

- Take a large, plastic syringe and pull back the plunger.
- Insert the nozzle into a disk of raw potato so a small piece is wedged inside the nozzle.
- Aim it away from yourself and other people and quickly depress the plunger.

(L) *Describe what happens.* _____

(M) *Explain this result.* _____

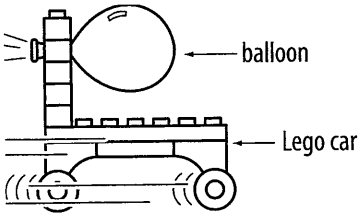
(N) *Many movies and television programs have staged gunfights. They often show someone who has been shot being knocked backwards by a metre or more, while the shooter hardly experiences any recoil. Use your knowledge of Newton's third law (for every action there is an equal and opposite reaction) to discuss this.* _____

_____

13.5 Balloon power

A balloon can be made to propel a toy car.

Materials: Lego kit with wheels, balloon, sticky tape



Procedure

- Use the illustration to help you design and make a balloon-powered vehicle.

(O) *What is the maximum distance you can get it to travel?* _____

(P) *Blowpipes are traditional weapons in parts of Malaysia. Using only their lung power, hunters can blow a dart over 100 metres. Give reasons why hunters might prefer to use blowpipes rather than rifles to shoot their prey.* _____

_____

(Q) *Two people of equal mass are on roller skates, facing each other, when one skater pushes the other. Use Newton's third law to describe what would happen.* _____

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14_SWINGING

The pendulum is a familiar object in science laboratories. A pendulum consists of a mass, attached by a cord to a static point. Once the mass is set swinging, it will oscillate (move back and forth in a regular manner). Galileo Galilei was the first person to record the period (time to swing 'there and back') around the year 1583. The following all have some similarities to pendulums.

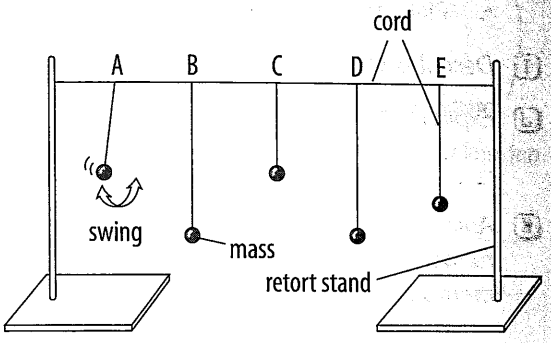
14.1 The more the merrier T

This experiment shows you a way of moving kinetic (moving) energy.

Materials: two stable supports (e.g. retort stands clamped to the bench), strong cord, masses (mass carriers optional)

Procedure

- Attach five masses to cords.
- Two cords (A and C) should be the same length.
- Two other cords (B and D) should be the same length, but longer than A and C.
- One cord (E) should be intermediate in length between the two others.
- Another cord should be tied tautly between two retort stands.
- Make sure all masses are steady.
- Start mass A swinging, as shown.
- Leave the apparatus for a few minutes.



(A) *Describe what happens to the system.* _____

- Steady all the masses.
- Start mass B swinging.
- Leave the apparatus for a few minutes.

(B) *Describe what happens to the system.* _____

- Repeat with other masses, one at a time.

(C) *Describe what happens.* _____

(D) *Comment on your results.* _____

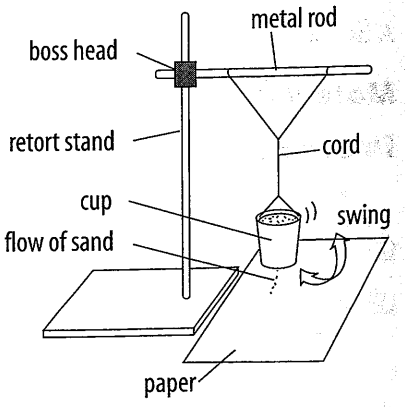
14.2 Lissajous figures (Bowditch curves) T

In 1857–58 Jules-Antoine Lissajous (a French mathematician) experimented with a device using the same principles as the one shown below. This is a compound pendulum, which has a cup that dribbles sand as it moves. A compound pendulum is one that swings between two axes set at right angles to each other.

Materials stable support (such as a retort stand clamped to a bench), cord, metal rod, boss head, paper (or polystyrene) cup, fine dry sand, pencil, sheet of paper

Procedure

- Attach a paper cup to cords as shown.
- Ensure that there is a sheet of paper spread under the paper cup.
- Fill the paper cup with sand.
- Punch a hole in the bottom of the paper cup, just sufficient for a thin, slow flow of sand.
- Stop the flow of sand with your finger.
- Start the cup swinging slowly and remove your finger.
- Ensure that the flow of sand doesn't go outside the paper.



(E) *Describe what you see.* _____

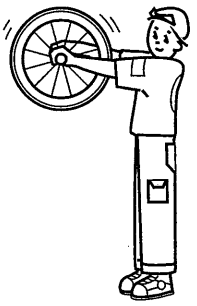
- Repeat, using different lengths of cord.

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15_ALL IN A SPIN-A

Although we tend to think of movement as being in a straight line from point A to point B, there are many other ways that objects move. If an object moves about a central point (axis), then it has circular motion. Like objects moving in straight lines or stationary objects, they still have inertia.

15.1 Wheels keep on turning



A spinning bicycle wheel seems to have a mind of its own.

Materials: a range of bicycle wheels, two pairs of pliers (optional), two volunteers

Procedure

- Hold the wheel in the front of you and use pliers to grip each end of the axle.
- Ask another person to spin the wheel vigorously.
- While the wheel is spinning quickly, turn to face another direction.

(A) *Describe what happens.* _____

(B) *Explain why the wheel behaves as it does.* _____

_____

- Repeat the investigation, varying the speed of the spinning wheel.

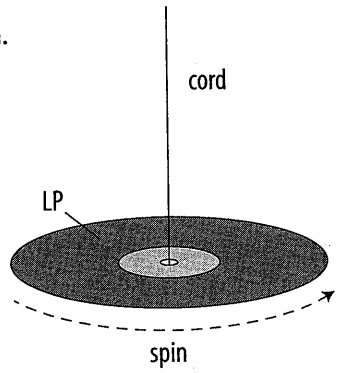
(C) *Describe what happens.* _____

- Repeat the experiment using wheels of different sizes.

(D) *Describe what happens.* _____

15.2 Spinning a disc

A twelve-inch (30 cm) LP (long play) record can be used to illustrate inertia in circular motion.



Materials: LP record, cord, matchstick

Procedure

- Tie a a piece of cord around a matchstick.
- Thread the cord through the hole in the LP until the matchstick supports the LP.
- Swing the LP as a pendulum.
- Steady the LP.
- Start the LP spinning as shown, then try to swing the LP.

(E) *Describe what happens.* _____

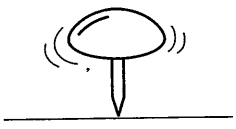
(F) *Explain why the LP behaves as it does.* _____

15.3 Spin a drawing pin

Materials: drawing pin, tabletop

Procedure

- Carefully hold the spike of a drawing pin.
- Give it a sharp twist and let it drop onto the tabletop.



(G) *Describe what happens.* _____

(H) *Explain the pin's behaviour.* _____

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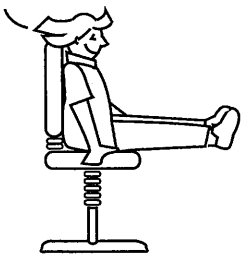
15.4 Spinning chairs

The momentum of an object moving in a straight line is its mass multiplied by its speed. Rotating objects have a similar property called angular momentum. To get the angular momentum of a rotating object, you multiply its moment of inertia by its angular velocity. The moment of inertia depends on the mass of the object and how this mass is distributed. Angular velocity depends on how rapidly the object is rotating.

You can illustrate conservation of angular momentum using an office chair that spins.

! SAFETY NOTE !
Make sure this is done in a clear area.

Materials: office chair that spins, open area, volunteer



Procedure

- Sit on an office chair that spins.
- Start the chair spinning with your legs stretched out in front of you.
- Then draw your legs in close to your body.

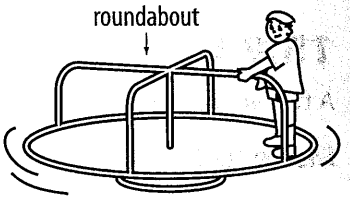
(I) *Describe what happens.* _____

(J) *Complete the passage to explain what you observed. Use these words: angular, conserved, decreases.*

Because angular momentum is _____ *then the moment of inertia multiplied by angular velocity must remain the same. So if one of these quantities* _____, *then the other one must increase and vice versa. When you pull in your legs, the moment of inertia decreases so* _____ *velocity must increase.*

15.5 Moving in a roundabout way

Einstein was famous for his 'thought experiments'. You can carry out a 'thought experiment'. Imagine the following. You go to a playground. You stand near the edge of a roundabout and start it moving slowly. After this you move to the centre of the roundabout and hold on tightly.



(K) *Describe what would happen.*

_____

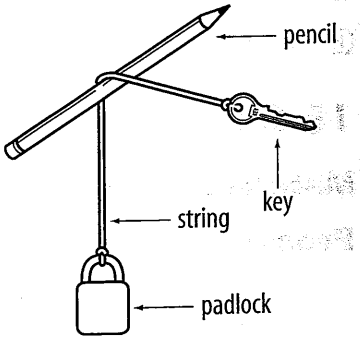
(L) *What is another sport or activity that involves spinning and a change of speed when arms are moved out or in?*

_____

15.6 Falling padlock

The pull of gravity on a padlock can be turned into circular motion.

Materials: pencil, key, padlock, scissors, metre rule



Procedure

- Cut a piece of string about 1 metre long.
- Tie one end to a heavy padlock and the other to a small key.
- Hold the key and use a pencil to support the string about halfway down its length, as shown.
- Let go of the key.

(M) *Describe what happens.* _____

(N) *Explain why the key behaves as it does.* _____

_____

Investigation

(O) *What would happen if the key was held closer to the pencil? Or if the positions of the key and the padlock were reversed?*

(P) *In the time it takes you to jump into the air and land again, the ground underneath you has moved over 200 metres eastward, due to the Earth's spin. Why do you land in the same spot?*

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16_CENTRIPETAL FORCE-A

A special force is needed to keep objects travelling in a circle.
This is an inward acting force called centripetal force.

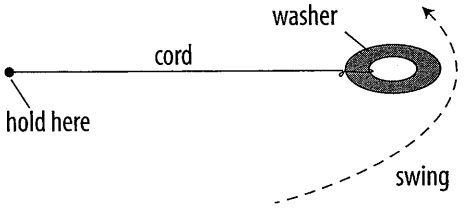
! SAFETY NOTE !

Make sure you are in a clear area and no one will be in range of the washer.

16.1 Going in circles

This simple experiment will help you understand centripetal force.

Materials: short length of cord, washer



Procedure

- Securely attach a short piece of cord to the washer.
- Swing the washer around your head as shown.

(A) *What do you feel on the fingers that hold the cord?*

_____

(B) *What is the name of the force that you exert to keep the washer and cord moving as they do?* _____

(C) *What would happen if the washer suddenly became detached from the cord?* _____

16.2 Keys that lift method 1 (Teacher demonstration only)

You can use some keys (and a bit of circular motion) to lift an object of much greater mass.

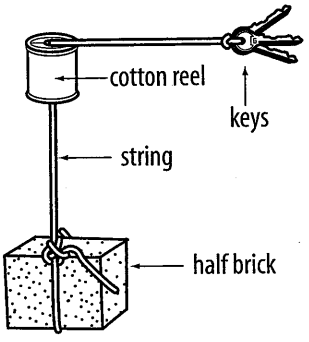
! SAFETY NOTE !

- Do this activity outside in a position where students can observe without risk of injury from the fast moving keys.
- Hold the cotton reel above your head so the keys won't hit you in the face.

Materials: three large keys, strong string, cotton reel, scissors, half brick, metre rule

Procedure

- Cut a one-metre length of string.
- Securely tie three big keys to one end of the string.
- Thread the rest of the string through the hole in a cotton reel and tie the other end to half a brick (or an object with similar mass).
- Hold the cotton reel and use it to very vigorously spin the keys in a circle.



(D) *What happens to the brick?* _____

Explanation

The force you have given to the keys would actually make them move in a straight line away from you if the string was cut. (Due to their inertia, they would resist a change in speed and direction and thus tend to move at a tangent to the spinning circle.) However, the inward-acting centripetal force exerted by the brick, via the string, prevents this from happening. When you spin the keys very quickly, the inertia overcomes the centripetal force, and the brick is lifted.

16.3 Keys that lift method 2

You can investigate the principles looked at in the previous experiment using much lighter equipment. Imagine you are supplied with some thread, washers, paper clips and the tube from a ballpoint pen.

(E) *Outline how you could carry out this experiment.* _____

_____

(F) *Draw the assembled apparatus in the box provided.*

(G) *Outline the results you might expect to obtain.* _____

_____

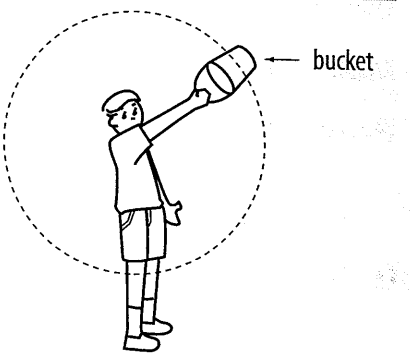
(H) *True or false? The Earth's spin is gradually slowing down.* _____

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16.4 Swing a bucket of water

! SAFETY NOTE !
Do this activity outside and stand well clear of the bucket.

If you tip a bucket of water upside down, the water will fall out. Using the technique described below, you can keep water in a bucket even though it has the open end above your head.



Materials: small plastic bucket, water, volunteer

Procedure

- Add water to the bucket until it is about three-quarters full.
- Begin swinging the bucket smoothly until you can swing it over your head in a big circle. (If you swing it quickly enough the water will remain in the bucket.)

(I) *After a class discussion, outline the science behind this experiment.*

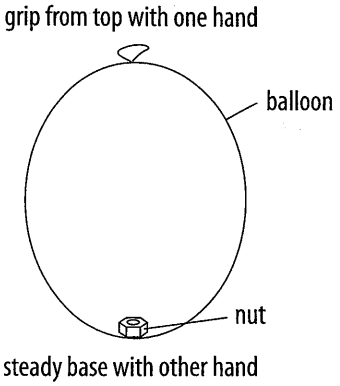
16.5 Nut and balloon

! SAFETY NOTE !
Make sure the nut is not swallowed.

A balloon and hexagonal nut can help you understand inertia and centripetal force.

Materials: large latex balloons, small (6 mm or 1/4") hexagonal nut, five-cent coin

Procedure



- Push the nut though the neck of the balloon, ensuring it goes completely through.
- Blow up the balloon and tie the neck.
- Grip the balloon from the top with all five fingers of one hand.
- Rotate the balloon until the nut starts to roll around the inside of the balloon.
- Use the other hand to stabilise the lower part of the balloon.

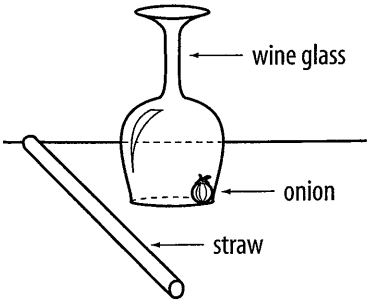
(J) *Describe what happens.* _____

- Repeat the experiment, replacing the nut with a five-cent coin.

(K) *Describe what happens.* _____

16.6 Know your onions

Here's another 'thought experiment' that can be tried at home. Imagine the following. Place a straw on a table and a small cocktail onion to one side of it. The problem is: how can you use an upside down wineglass to lift the onion over the straw to the other side without touching the onion with your hands?



(L) *Give your solution and explain why it would work. (Hint: use circular motion.) One solution is below – in 'upside down' form.*

It can be done by placing the wineglass over the onion, then swirling the glass so the onion rolls around the inside surface. If you can get the onion moving quickly enough, it will rise up the inside of the wineglass until it is rolling around the widest point. The glass can then be lifted over the straw to deposit the onion on the other side.

Research

(M) *Centrifugal force is a controversial term. Engineers use it in their calculations, but physicists say it doesn't exist. In your workbook, outline what is meant by centrifugal force.*

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17_BUMPING INTO THINGS

Energy is exchanged when objects collide with (bump into) each other.

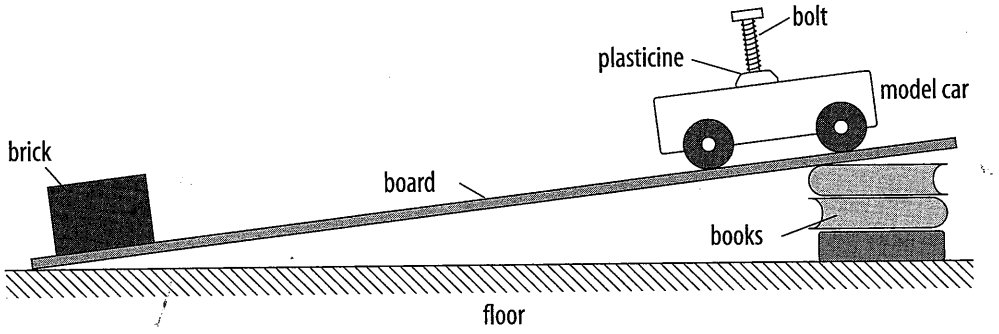
17.1 Crumple zones T

It is an unfortunate fact of life that cars sometimes have collisions. When cars suddenly stop, the kinetic (moving) energy can be transferred to the occupants of the car. There are 'crumple zones' in the front and rear of modern cars. These are areas that are designed to crush easily on impact, to absorb energy and prevent the full force of the collision being passed onto the occupants. The bolt in this experiment represents the occupants of a car. The more the bolt moves, the more the people inside would be affected by the crash.

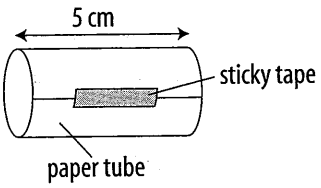
Materials: model car (dynamics trolley), plasticine, protractor, sticky tape, rubber bung, paper, card, scissors, retort stand, textbooks, length of wood (approx 1 m), bolt, brick

Procedure

- Attach a piece of plasticine to the top of the 'car'.
- Stick the bolt in the plasticine so that it is standing up at a 90-degree angle to the roof of the car.
- Arrange the board as shown. (The slope should be great enough to allow the car to build up some speed.)



- Allow the car to roll down the hill to where it will be stopped by the brick.
- Use a protractor to measure how many degrees the bolt moved.
- Reinstate the bolt to its original position.
- Attach a rubber bung to the front of the car using sticky tape.
- Repeat the experiment.
- Cut a piece of paper 5 cm by 10 cm.
- Make it into a tube as shown (illustration not to scale).
- Attach your tube to the front of the car.
- Repeat the experiment.
- Replace the paper tube with one made from a 10 cm by 10 cm piece of paper.



(A) *Record your results in the table below.*

Attachment	None	Rubber bung	5 cm x 10 cm paper tube	10 cm x 10 cm paper tube
Movement of bolt (°)				
Observations				

(B) *Which version would protect the car's 'passengers' best? Explain your answer.*_____

_____

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18_BRIDGING THE GAP

Bridges are structures used by people (and vehicles) to get over obstacles.

Ⓐ *List two obstacles that bridges can cross.* _____

Ⓑ *Bridges can be made of many types of materials, such as simple beams of wood, bricks and concrete. Give one other material used to build bridges.* _____

A bridge needs to be strong enough to safely support its own weight as well as the weight of the vehicles and people that pass over it.

Ⓒ *What else must a bridge be designed to withstand?* _____

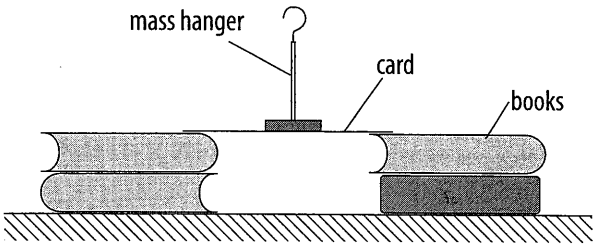
You can model how large structures work using simple materials.

18.1 Beam bridges

The simplest type of bridge is a beam.

Materials: thin card, ruler, scissors, masses, mass hangers, textbooks

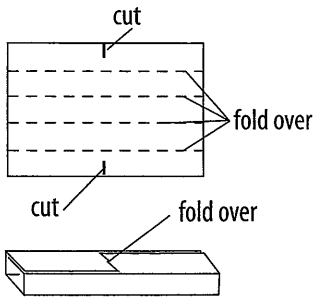
Procedure



- Cut A4 card into 4 equal-sized pieces (approx. 15 cm x 10.5 cm each).
- Place two piles of books 12 centimetres apart.
- Bridge the gap with the card.
- Add masses to the card until it bends.
- Note the mass added.
- Fold the card into pleats as shown.
- Bridge the gap with the folded card.
- Add masses until the card bends.
- Note the mass added.

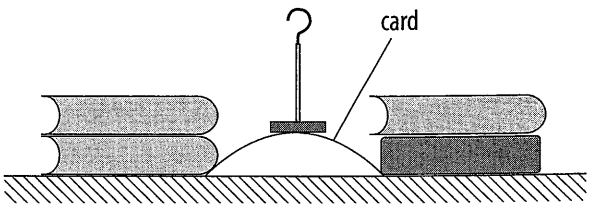


- Divide a card lengthways into five equal parts.
- Cut outside sections at centre halfway into outside sections, as shown.
- Fold and form a beam as shown.
- Bridge the gap with your folded card.
- Add masses until the card bends.
- Note the mass added.



18.2 Arch bridge T

- Make an arch with card and place it between books as shown. Books should be 12 cm apart.
- Add masses until card bends.
- Note mass added.



Ⓓ *Record your results in the table below.*

Type of bridge	Mass needed to bend card (g)

Ⓔ *Which type of bridge was the strongest?* _____

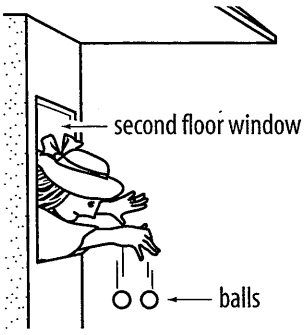
Research

Ⓕ *Find out how suspension bridges work.*

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19_A MATTER OF GRAVITY

Gravity on Earth is a force that pulls all objects towards the centre of the Earth. Two scientists who did very important research on gravity (and astronomy) were Galileo Galilei (1564–1642) and Sir Isaac Newton (1643–1727). Galileo is credited with being the first person to investigate the speed at which objects fall, and there is a famous story of him dropping balls from the Tower of Pisa. (Although there is no factual basis for this anecdote.) You too can investigate how objects fall under the influence of gravity.



19.1 Drop everything!

Materials: balls of same size but different weights, second-floor window, open space

Procedure

- Organise the class into 'droppers' and 'observers'.
- Droppers take the balls to the second-floor window.
- Observers station themselves where they can see the balls being dropped, but are not in danger of being hit by the balls.
- Droppers drop the balls all at the same time.
- Observers check when the balls reach the ground.

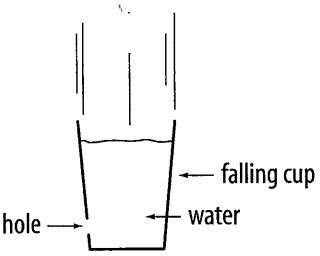
! SAFETY NOTE !
Ensure that the balls are soft and no one is in danger of being hit by falling balls. Do not use golf balls.

(A) *Do the balls differ in the time they take to reach the ground?* _____

19.2 In free fall

How can you keep water in a cup with a hole in it?

! SAFETY NOTE !
This activity should be done outside.



Materials: polystyrene cup, pen, water

Procedure

- Fill a polystyrene cup with water.
- Use a pen to put a hole near the bottom of the cup. (The water will drain out.)
- Place your finger over the hole and fill the cup again.
- Hold the cup up high and let it drop to the ground.

(B) *Describe what happens.* _____

(C) *Underline the correct words in italics to provide an explanation for what happened to the cup.*

The force of *gravity/friction* makes the water drain from the cup when it is held in the air. When the cup is *falling/rising*, both the water and the cup are *accelerating/decelerating* at the same rate so the water is *actually/apparently* weightless in comparison to the cup.

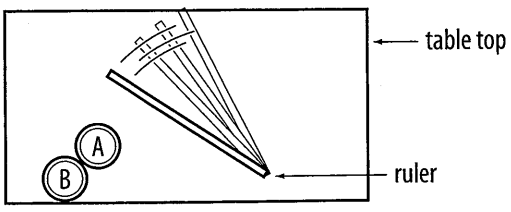
19.3 Colliding coins

Does the path of an object influence the time it takes to fall to the ground?

(D) *After reading the procedure, but before you carry out the experiment, forecast what will happen to the coins.* _____

_____

Materials: 2 x five-cent coins, ruler, table



Procedure

Hint: you will hear the sound better on a wooden floor.

- Place two five-cent coins, touching each other, near the edge of a table in the positions shown.
- Flick coin A with a ruler, so it is propelled across the room, and coin B will fall more directly to the floor.

(E) *Describe what actually happened and if it matched your forecast.* _____

(F) *If a bullet is fired from a gun that is horizontal, it will hit the ground at the same time as another bullet which is simply dropped from that height. Explain why this happens.* _____

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20_DOING THINGS MECHANICALLY-A

A machine is a device that changes force or motion (movement). Machines transfer energy from place to place. Machines make work easier; they do not do less work nor use less energy. A pulley is a machine.

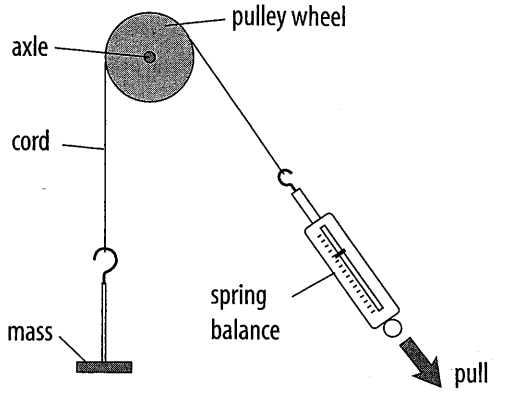
20.1 What does a pulley do? T

A pulley is a wheel with a groove to carry rope. The rope is pulled at one end (effort) to lift a weight (load) at the other end.

Materials: pulley wheel, support, cord, spring balance, mass, ruler, attachment point (e.g. retort stand and clamp)

Procedure

- Set up the apparatus as shown. You will have to support the pulley by the axle (e.g. by using a retort stand, boss head and clamp).
- Ensure that the mass and spring balance are stationary.
- Note the value of the mass.
- Note the reading on the spring balance.
- Pull the mass upward using the spring balance.
- Repeat using three different masses.



(A) *Record your results in the table below.*

Mass (g)	Spring balance (g)

(B) *How do the readings on the spring balance and masses compare?* _____

(C) *What is the purpose of a pulley?* _____

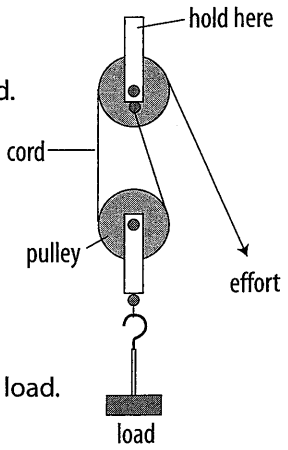
20.2 Pulling power T

A block and tackle is made up of at least two pulleys and can be used to lift heavy objects using a small amount of force. Archimedes (the Greek philosopher) is reputed to have invented the block and tackle around 2200 years ago. The illustration shows a diagram of a two-pulley block and tackle. Now we will investigate the relationship between the effort you put in to lift an object and the mass of the object, and also the distance moved by effort and by the load.

Materials: cord, two pulleys, load (masses), force meter, attachment point (e.g. retort stand and clamp), metre rule

Procedure

- Attach the top pulley to a secure point.
- Attach load (masses) to the bottom pulley and record the weight of the mass attached.
- Attach the cord to the bottom of the upper pulley.
- Loop the cord through the bottom pulley.
- Pass the cord over the top pulley.
- Attach the force meter to the cord.
- Pull on the force meter to lift the load
- Note the reading on the force meter.
- Compare the distances travelled by the force meter with the distance travelled by the load.
- Repeat with different loads.



(D) *Draw up a table in your science book to record your results.* _____

(E) *Conclusion* _____

(F) *Forecast what would happen if you used a block and tackle with four pulleys.* _____

_____

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20.3 The power of one

You can make a block and tackle from two broomsticks and a length of rope.

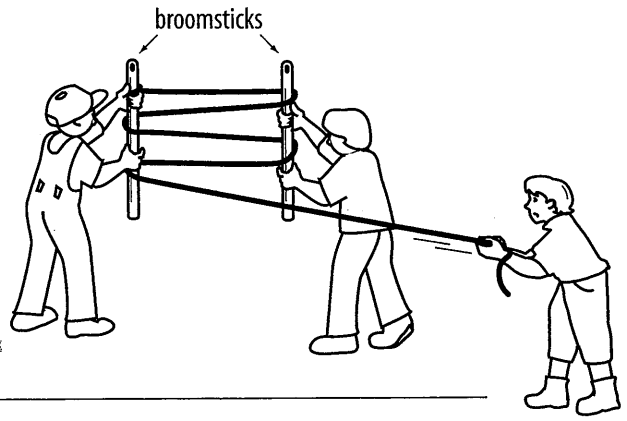
! SAFETY NOTE !

Fingers should be kept clear of the rope. This should be done in a clear area.

Materials: two broomsticks, length of rope (about ten metres)

Procedure

- Get two volunteers to each hold a broomstick, about one metre apart.
- Tie the end of a rope to one of the sticks, then wind the rope three times around both sticks, as shown.
- Ask a third person to pull the rope while the other two try to stay where they are.



(G) *Describe what happens.* _____

_____

The sticks and rope shown in the illustration are like a block and tackle with four pulleys. Theoretically the puller should only need one quarter of the force normally needed to pull the other two students together. However, in the activity above, the efficiency of the system is reduced.

(H) *Give one reason why this pulley system is not 100 per cent efficient.* _____

_____

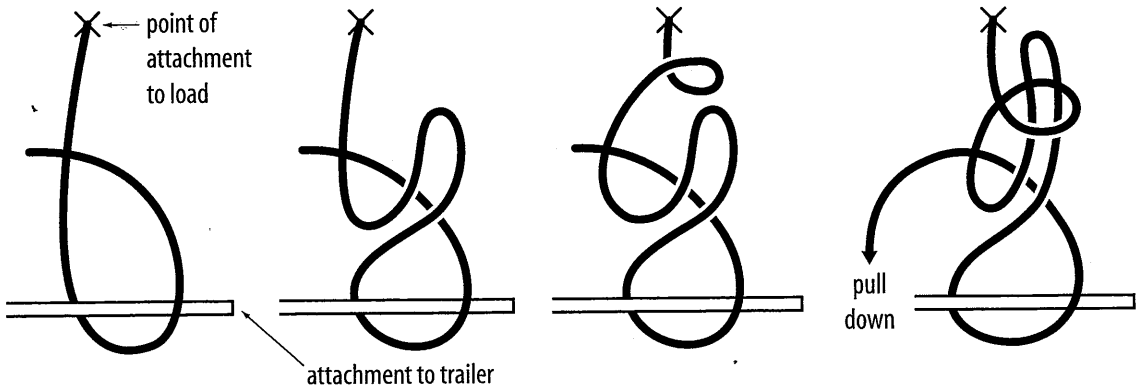
20.4 Knot hard?

A 'truckie's knot' is shown in the illustration. It is a block and tackle completely made of rope. It is useful for tying down loads on trucks and trailers.

Materials: cord or rope, two points of attachment

Procedure

- Attach one end of the rope to a load.
- Tie the knot as shown.



(I) *Discuss the advantages and disadvantages of using a truckie's knot rather than pulleys.* _____

_____

Research

(J) *There are many prehistoric buildings that have huge blocks of stone that have been raised to great heights. These include the pyramids of Egypt, the statues on Easter Island and Stonehenge in England. Scientists still do not agree how the large masses of stone were moved. For one of these buildings, outline a theory explaining how these constructions came about.*

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21_A FRICTION STORY

When two surfaces move over each other, friction is the force that resists motion. Friction acts in the opposite direction to motion and depends on several factors. One of these is the roughness of the surfaces in contact. Friction can generate heat when objects are moving – the greater the friction the greater the heat.

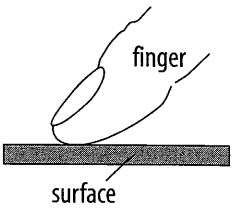
21.1 Taking the rough with the smooth T

You can see the effects of friction with different surfaces very easily.

Materials: bench, book, sandpaper and other surfaces

Procedure

- Press your finger on the bench.
- Rub the finger backwards and forwards for 15 seconds.
- Repeat with other surfaces.



(A) *Which surface generated the most friction?* _____

(B) *How did you come to your conclusion for (a)?* _____

(C) *Why did the surface mentioned in (a) create the most friction?* _____

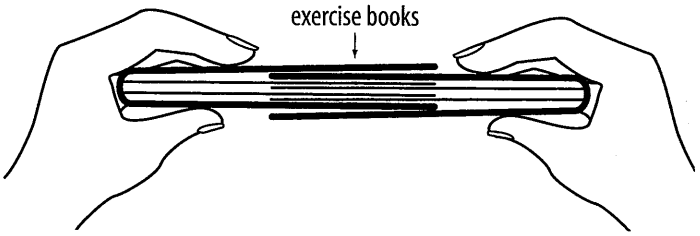
21.2 Books need to exercise!

Friction resists movement. The greater the area of contact between surfaces, the greater the friction.

Materials: two exercise books

Procedure

- Take two exercise books and interlock them by alternating at least 20 pages from each book.
- Make sure they overlap at least half of their width.
- Hold the books at the binding.
- Try to pull the books apart.



(D) *Describe what happens.* _____

(E) *Explain your result.* _____

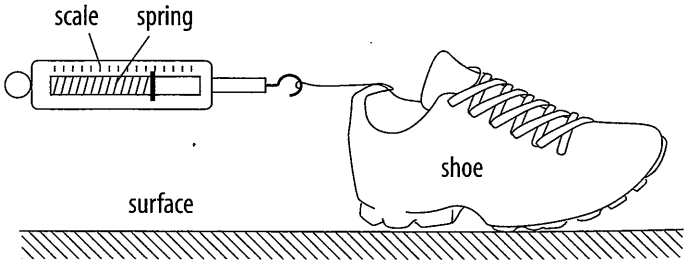
21.3 What a drag!

A force meter measures forces; usually in newtons (N). Most force meters have a spring and a scale. A force meter can help measure friction. You can investigate the grip of a sports shoe by measuring the force needed to drag it.

Materials: force meter, sports shoes, surface

Procedure

- Place a shoe on a surface and attach the force meter.
- Drag the shoe at a steady speed. (Be careful not to over-extend the spring in the meter.)
- Note the steady reading on the force meter.
- Repeat with other shoes and surfaces.



(F) *Record your results in a table in your workbook.*

(G) *Outline the factors that could contribute to the different amounts of force needed to drag the shoes.* _____

_____

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22_MOBILE MACHINES

Machines are devices that help you do things more easily. They can also change one form of energy to another. Study the following types of energy and state another name for each type.

(A) *Kinetic energy* _____

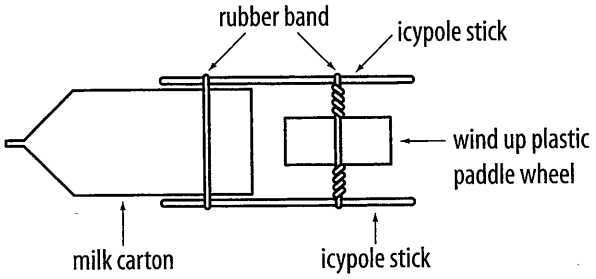
(B) *Potential energy* _____

22.1 Paddle boat

Materials: empty milk carton, stapler, water, two wooden icy pole sticks, two rubber bands, piece of flat plastic, tub of water

Procedure

- Add water to the milk carton until it floats with about half of the carton submerged when it is placed in water.
- Staple the flaps of carton closed.
- Assemble the paddleboat as shown.
- Wind up the plastic paddle wheel.
- Place the paddleboat in a tub of water.
- Release the plastic paddle wheel.



(C) *Describe what happens.* _____

(D) *What type of energy was put into the rubber band as it was wound up?* _____

(E) *What type of energy was in the rubber band when it was finally wound up?* _____

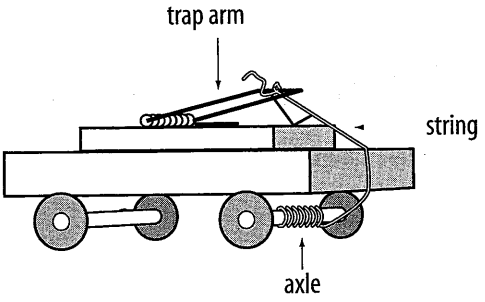
(F) *What type of energy was given out by the rubber band when it unwound?* _____

(G) *Is the paddle boat you made a machine? Explain your answer.* _____

_____

22.2 Mobile mouse trap

The illustration shows how a mechanical spring mousetrap can be used as a power source. One end of a length of string is attached to the spring lever of the mouse trap and the other end is wound around the front axle of the car.



(H) *Explain how this device would work. Include the energy changes involved.*

_____

_____

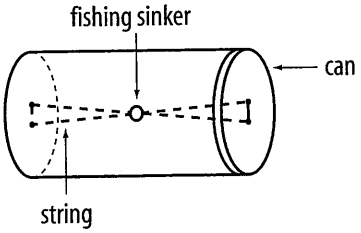
22.3 Many happy returns

You can make a simple machine that moves in an unexpected way.

Materials: can with a lid, nail, thin string (or fishing line), heavy fishing sinker, hard level surface

Procedure

- Make two nail holes about 1 cm apart in the base of the can, close to the centre.
- Repeat this with the lid.
- Attach a heavy fishing sinker to the centre of the string.
- Connect the holes with the string, as shown.
- Gently roll the can on a hard, level surface.



(I) *Describe what happens.* _____

(J) *Explain why the tin behaves as it does. Include any energy changes.* _____

_____

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23_FEELING THE PRESSURE?

It is not obvious, but air has mass and presses down on us. We can carry out some experiments to show what it does.

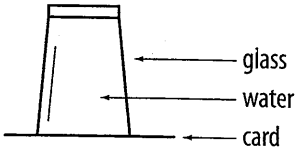
23.1 Card and glass

Materials: glass, playing card (or piece of card), water

! SAFETY NOTE !
This is best done over a sink, in case of mishaps.

Procedure

- Fill a glass with water.
- Put a playing card over the top. The card should completely cover the top of the glass.
- Put your finger on the card to hold it in place.
- Turn the glass upside down.
- Remove your finger from the card.



(A) *Describe what happens.* _____

(B) *Outline the role played by air pressure in this experiment.* _____

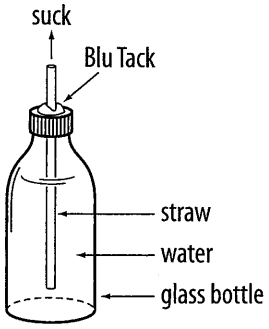
23.2 Are you a sucker?

This experiment will help you understand more about ambient air pressure. Air tends to flow from an area of high pressure to lower pressure. We use a straw to drink by sucking and reducing air pressure in the straw. Atmospheric pressure forces the drink up the straw.

Materials: clean glass bottle (with lid), straw, water, Blu Tack

Procedure

- Fill the bottle with water.
- Make a hole in the glass bottle's lid and insert a straw through it.
- Pack around the hole with Blu Tack to make it airtight.
- See how much water you can drink with one big suck.



(C) *How successful were you?* _____

(D) *Explain your result.* _____

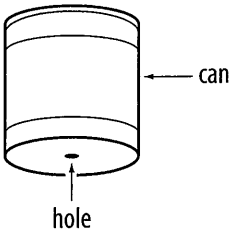
23.3 Can you pour water from one hole?

Materials: can with lid, tape, water, nail

! SAFETY NOTE !
This is best done over a sink, in case of mishaps.

Procedure

- Put a small nail hole in the bottom of an empty can.
- Temporarily block the hole with a piece of tape.
- Fill the can with water and put the lid on tightly, then remove the tape.



(E) *Describe what happens.* _____

- Open the lid.

(F) *Describe what happens.* _____

(G) *Discuss the role air pressure plays in this experiment.* _____

(H) *It is difficult to swat a housefly because it is extremely sensitive to changes in air pressure. As a rolled up newspaper descends towards the fly, it can sense the wind of its movement in time to escape. Why might a fly swatter with holes in it be more successful than a rolled up newspaper in swatting flies?* _____

Research

(I) *Air pressure is measured using a barometer. Find out the units used to measure air pressure.* _____

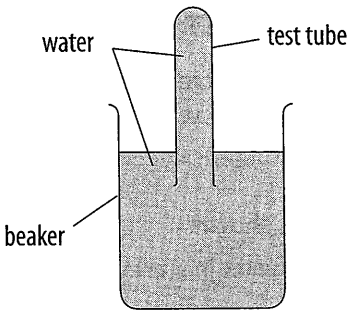
(J) *True or false: Canned food was available by about 1820, but the can opener was not invented until more than 40 years later. In the meantime, people used hammers and chisels to open their cans.* _____

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24_KEEPING UP THE PRESSURE

Air pressure can exert force.

24.1 Holding up water



Materials: beaker, water, test tube

Procedure

- Two-thirds fill a beaker with water.
- Place a test tube in the beaker.
- Fill the test tube with water.
- Invert the test tube.
- Lift the test tube upwards until its mouth is just under the level of the water in the beaker.

(A) *Describe what you see.* _____

(B) *Air pressure is responsible for your results. On the illustration, use arrows to show how air pressure and the force exerted by the water in the test tube act on the system.*

24.2 Breaking a rule (Teacher demonstration only)

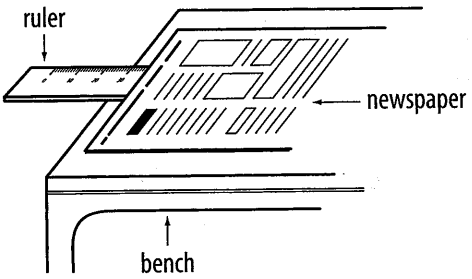
The greater the surface area of an object, the greater the air pressure on it.

Materials: light wooden ruler, newspaper, bench

! SAFETY NOTE !
Wear safety glasses.

Procedure

- Place a light wooden ruler on a bench so that about one-third of its length is projecting out from the edge of the bench.
- Place a spread-out newspaper over the part of the ruler that is on the bench.
- Hit the exposed end of the ruler very firmly.



(C) *Describe what happens.* _____

_____

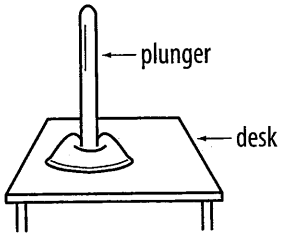
(D) *On the illustration, show where the forces on the ruler are acting when it is hit. (Hint: don't forget air pressure.)*

(E) *Explain why the newspaper must be spread out for the ruler to be broken.* _____

24.3 Taking the plunger

The 'plumber's helper' (plunger) is a flexible rubber cup attached to a wooden handle. It is most often used to unblock sinks. This experiment shows a different use.

Materials: plunger, desk, water



Procedure

- Wet the rubber on the rim of the plunger.
- Place plunger on the desk as shown.
- Pull the plunger upwards.

(F) *Describe what happens.* _____

(G) *Explain the role played by air pressure in this experiment.* _____

_____

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25_RELIEVING THE PRESSURE

25.1 Feeling crushed? (Teacher demonstration only)

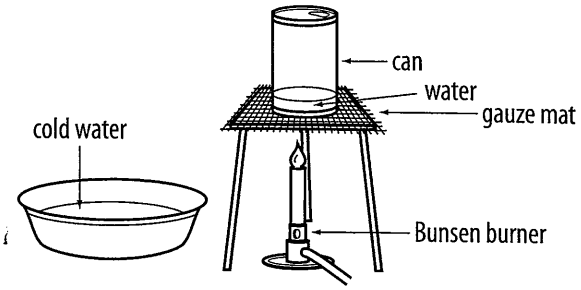
Air pressure exerts a considerable force (about 1 kg of pressure per square centimetre) and its effects can be surprising.

Materials: sink or basin full of cold water, Bunsen burner, gauze mat, empty soft drink can, water, metal tongs

! SAFETY NOTE !
Wear safety glasses and lab coats.

Procedure

- Set up the apparatus as shown.
- Boil a small amount of water in a used soft drink can.
- When the water is boiling and steam has displaced all of the air from the can, use a large pair of tongs to quickly lift the can towards the basin of cold water.
- Invert the can when it is over the cold water and immediately dip the top of the can downwards into the cold water.



(A) *Describe what happens.* _____

(B) *Explain why the can behaves as it does.* _____

_____

Calculation

We can work out the total air pressure on the can. The first step is to calculate the surface area of the can.

(C) *Measure the diameter and height of the can: Diameter =____ cm and height =___ cm.*

(D) *Surface area = (2 x 3.142 x 0.5 x diameter2) + (3.142 x diameter x height) =____ cm².* _____

(E) *Calculate the total air pressure. Assume that air pressure is 1 kg per cm². Show your working.* _____

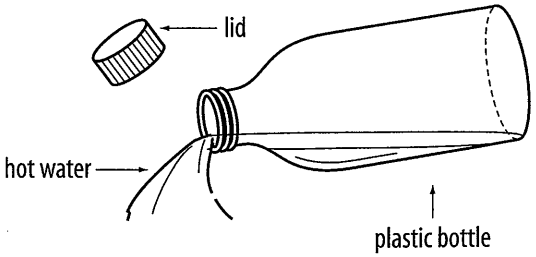
25.2 Bottle it up

Air expands when it is heated and contracts when it is cold.

Materials: plastic bottle, hot water, cold water

Procedure

- Pour some hot tap water into a plastic bottle.
- Swirl the water around inside the bottle.
- Tip the water out and immediately screw the lid on tightly.
- Run some cold tap water over the bottle.



(F) *Describe what happens.* _____

(G) *Explain why the bottle behaves as it does.* _____

25.3 Inside out inflation

Materials: flask, hot water, balloon

Procedure

- Rinse out a one-litre flask with hot tap water and empty it.
- Immediately fit a balloon over the mouth of the flask.

(H) *Describe what happens.* _____

(I) *Explain why the balloon behaves as it does.* _____

_____

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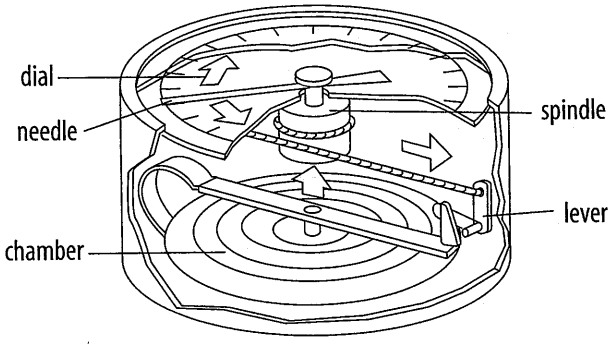
26_CHANGES IN AIR PRESSURE

The branch of science studying the processes of the Earth's atmosphere is called meteorology. By understanding the relationship between atmospheric pressure, temperature, cloud formation, winds and so on, meteorologists can make predictions about weather patterns that are likely to evolve. A barometer is a device that measures atmospheric pressure.

26.1 Whatever the weather T

Atmospheric pressure is a good indication of future weather. This experiment investigates the relationship between atmospheric pressure and the weather.

Materials: barometer, thermometer, daily newspaper, access to weather reports



Procedure

- Set up an area where the barometer can be studied for at least one week.
- Record the forecast from daily newspapers, radio and TV.
- Record the temperature and atmospheric pressure daily.
- Record other details, such as rain, clouds and sunshine.
- Repeat for at least one week.

(A) *Discuss the relationship between the weather and air pressure.* _____

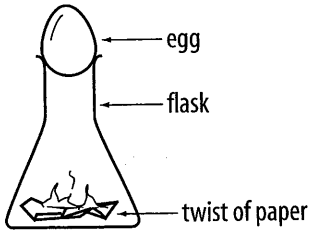
(B) *Discuss which source gave the most accurate weather forecasts, and the possible reasons for this.* _____

_____

26.2 Getting egged on (Teacher demonstration only) T

Hot air expands and cold air contracts. The next experiment makes use of this fact.

Materials: paper, matches, hard boiled egg, one-litre conical flask



Procedure

- Select an egg that is only slightly larger than the mouth of a one-litre conical flask.
- Hard boil the egg.
- Shell the egg and keep it wet.
- Drop a burning twist of paper into the one-litre conical flask.
- Place the wet egg in the mouth of the flask.

(C) *Describe what you see.* _____

(D) *Explain the part air plays in this experiment. Include a reason the flames go out.* _____

_____

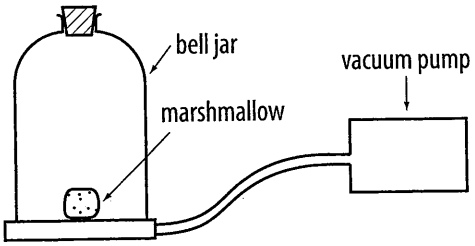
26.3 What would happen if ... ?

A marshmallow is a confection made up of a froth or foam. There are pockets of air inside marshmallows.

A vacuum pump removes the air from a sealed container.

Imagine the following

- A marshmallow is placed inside a bell jar.
- The bell jar is sealed and connected to a vacuum pump.
- The air is pumped out of the bell jar.



(E) *Describe what is likely to happen and the reasons for this.*

_____

_____

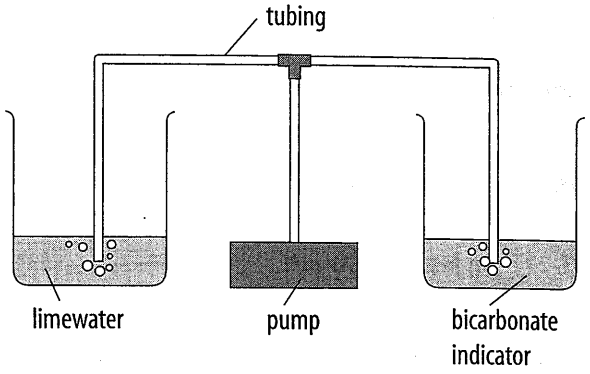
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27_IT'S A GAS

Carbon dioxide (CO_2) makes up around 0.03 per cent of the Earth's atmosphere by volume. The two main reagents used to test for carbon dioxide are bicarbonate indicator and limewater – a solution of calcium hydroxide, $Ca(OH)_2$. Carbon dioxide is an acidic gas and turns bicarbonate indicator from purple to yellow. When carbon dioxide is bubbled through limewater, the water goes from clear to cloudy. You can compare the sensitivity of these two chemicals to the presence of carbon dioxide.

27.1 Comparing chemicals T

Materials: aquarium pump, bicarbonate indicator, limewater, tubing, tube splitter/divider, beakers, stopwatch



Procedure

- Set up the apparatus as shown.
- Start the pump.
- Time how long it takes for both solutions to change in appearance.

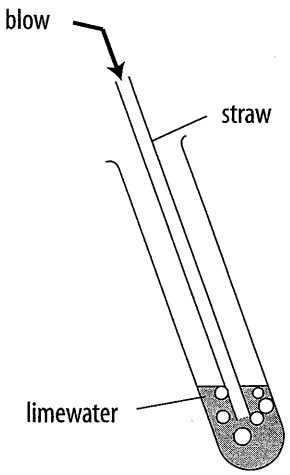
(A) *Limewater*_____ *Bicarbonate indicator*_____

(B) *Which solution was more sensitive to carbon dioxide? Explain your answer.* _____

_____

27.2 I'll be blowed

We breathe air in and then breathe it out again. This experiment looks at the amount of carbon dioxide in the atmosphere.



Materials: two test tubes, two straws, limewater, bicarbonate indicator, stopwatch

Procedure

- Place a few centimetres of limewater in a test tube.
- Place a straw into the limewater.
- Steadily breathe out through the straw to bubble your exhaled air through the limewater.
- Time how long it takes for the solution to change in appearance.
- Repeat with bicarbonate indicator.

(C) *Time taken for limewater to change:* _____

Time taken for bicarbonate indicator to change: _____

(D) *Which contains more carbon dioxide, the atmosphere or exhaled breath? Explain your answer.* _____

_____

In science investigations, for a true conclusion to be drawn, we must compare 'like with like', or in other words, only change one variable at a time.

(E) *Study the previous experiments involving testing for carbon dioxide. Describe any drawbacks in these, especially those that involve variables.* _____

_____

_____

Give the names of the compounds represented by the following formulae:

(F) CO_2 _____

(G) $Ca(OH)_2$ _____

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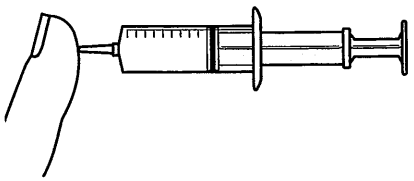
28_PUTTING PRESSURE ON AIR AND WATER

28.1 Compression of fluids

Air is a gas and water is a liquid. Can we compress these substances (i.e. make them take up less volume)?

Materials: plastic syringe, water

Procedure



- Pull back the plunger of a plastic syringe so that it is full of air.
- Place your thumb over the nozzle of the syringe (as shown) to seal it.
- Press the plunger to see if you can compress the air.
- Repeat the procedure using water in the syringe.

(A) *Describe what happens to the volume of the air.* _____

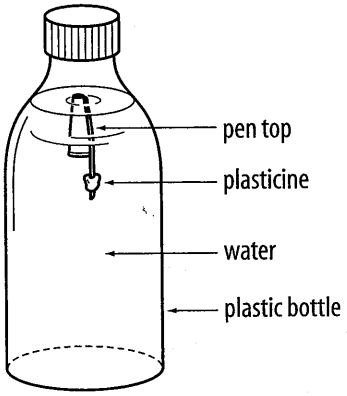
(B) *Describe what happens to the volume of the water.* _____

28.2 Cartesian diver

A floating pen lid can be made to sink and then rise without being touched.

Materials: plasticine, water, plastic bottle, plastic pen lid

Procedure



- Attach some plasticine to a plastic pen lid (as shown), so that it only just floats in water. If it has an opening at the top, seal this up, but be careful not to block the bottom opening of the pen lid.
- Lower the pen lid into a plastic bottle filled with water.
- Screw the bottle cap on tightly and then squeeze the sides of the bottle.

(C) *Describe what the pen lid does.* _____

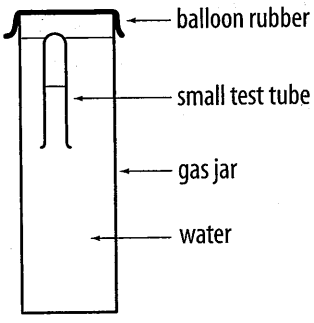
(D) *What happens to the volume of the air inside the pen lid when you squeeze the bottle? (You can observe it if you look into the pen lid from below.)* _____

Explanation

(E) *Use these words to complete the explanation: air, compressed, sinks, squeeze.*

When you _____ *the bottle, the water transfers the pressure to the* _____ *inside the pen top. The air inside the pen top becomes* _____, *so the unit (air pocket + pen lid) becomes more dense than the water, and it* _____.

(F) *The previous experiment showed one way to make a 'Cartesian diver'. The illustration on the right shows an alternative way to make a Cartesian diver using common laboratory equipment. Explain how you would set it up and cause the 'diver' to move up and down.*



_____

_____

Research

Are the following statements true or false?

(G) *Sperm whales can hold their breath for almost two hours.* _____

(H) *Sperm whales can swim down to one kilometre beneath the surface.* _____

(I) *Scuba divers breathe compressed air and may get 'the bends' if they come to the surface too quickly. Research the bends (decompression sickness). Is opening a bottle of fizzy drink similar to a scuba diver getting the bends? Explain your answer.*

Historical note The first submarine, built in 1620, was covered with leather and powered by oars. It travelled up the Thames River and could remain at a depth of 5 metres for about 15 minutes. Modern military submarines can circumnavigate the globe without surfacing.

(J) *What is meant by 'circumnavigate'?* _____

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29_FLUIDS UNDER PRESSURE

Hydraulics deals with the applications and effects of flowing liquids. Hydraulic systems are used to transfer power efficiently. These systems usually consist of water or oil under pressure and rely on the fact that liquids are (almost) incompressible.

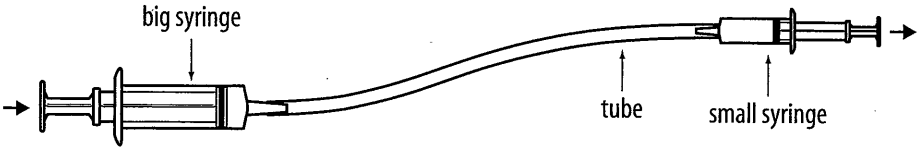
29.1 Syringe power

In the school laboratory plastic syringes and tubing can be used to investigate hydraulics.

Materials: large plastic syringe (with volume markings), small plastic syringe (with volume markings), ruler, water, plastic tube, scissors

Procedure

- Cut a thin, flexible piece of plastic tubing at least 50 cm long.
- Attach a small syringe to the tube.
- Fill the small syringe and tube with water.
- Attach a large syringe (with the plunger pushed in) to the other end of the tube.
- Note the volume of water in the small syringe.
- Press the plunger on the small syringe.



(A) *What does the plunger on the large syringe do?* _____

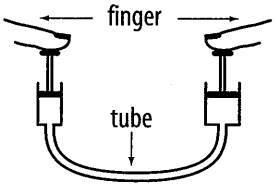
(B) *Measure and compare the volume change in both syringes.* _____

(C) *Measure and compare the distance moved by the plungers.* _____

29.2 When push comes to shove

If you have syringes with barrels of different diameters you can demonstrate how they can multiply force.

Materials: large plastic syringe, small plastic syringe, water, plastic tube



Procedure

- Set up the appparatus in the same way as for the last experiment.
- Place a finger on each plunger.
- Press downwards on the small plunger and try to resist the pressure on the big plunger.

(D) *Describe what happens.* _____

29.3 Bricks push up (Teacher demonstration only) T

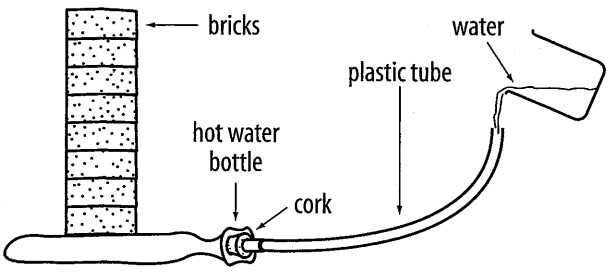
Hot water bottles are flexible rubber containers.

Materials: 8–10 bricks, hot-water bottle, cork with a hole, short piece of glass tubing, plastic tube, beaker of water

! SAFETY NOTE !
Stay well clear of the bricks.

Procedure

- Inset the glass tubing into the hole in the cork.
- Fit the cork into the opening of the hot-water bottle.
- Attach about 50 cm of plastic tubing to the glass tube and make sure all the fittings are secure.
- Stack a column of 8–10 bricks on top of the hot water bottle.
- Pour 200–300 mL of water into the tube.



(E) *Describe what happens.* _____

(F) *Explain your results.* _____

_____

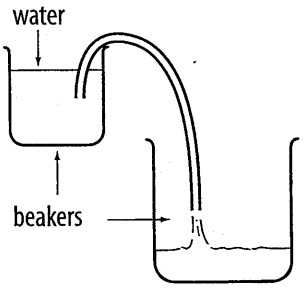
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30_WATER FALLS

Air pressure can affect how water flows.

30.1 Make a siphon



Materials: 200 mL beaker, 400 mL beaker, water, rubber tube

Procedure

- Hold a 200 mL beaker full of water on top of the bench so it cannot tip over.
- Position an empty, larger beaker below the bench.
- Place one end of a rubber tube in the water.
- Suck the other end of the tube and then immediately place that end in the lower beaker.

(A) *Describe what happens.* _____

(B) *On the illustration, show how air pressure causes the water to move.*

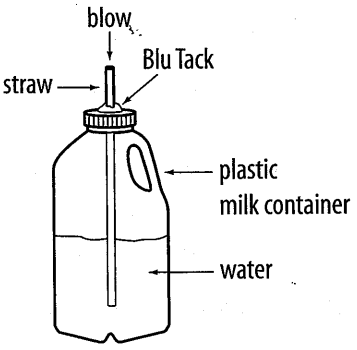
(C) *Someone wants to use a similar siphon to move petrol from one container to another. Outline possible problems with this.*

_____

30.2 A simple fountain

You can make a fountain using a few household objects.

Materials: empty, clean two-litre plastic milk container, plastic straw, water, Blu Tack



Procedure

! SAFETY NOTE !
This is best done in a sink to avoid mess.

- Put a small hole in the lid of the milk container.
- Push a strong plastic straw into the hole.
- Make the join airtight by lining it with Blu Tack.
- Half-fill the container with water.
- Put the lid on tightly.
- Blow strongly through the straw into the container, then stand back.

(D) *Describe what happens.* _____

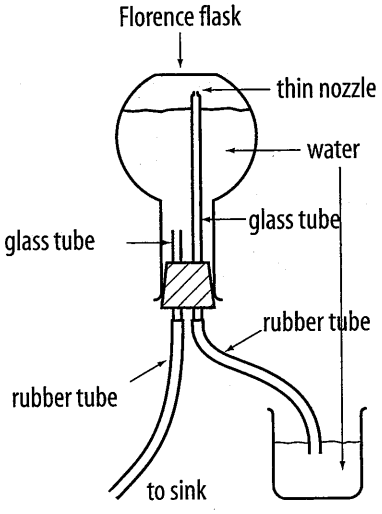
30.3 Water flowing uphill (Teacher demonstration only)

Materials: Florence flask (round-bottom), water, glass tubes, rubber tubes, beaker, rubber stopper, sink

! SAFETY NOTE !
The glass tubes should be inserted into the stopper by an experienced laboratory assistant because there is a risk of the glass breaking.

Procedure

- Add water to a Florence flask until the water level reaches the neck of the flask.
- Obtain a stopper with a long glass tube (with a narrow mouth) and a short glass tube already inserted.
- Attach rubber tubing to each one as shown.
- Insert the stopper into the mouth of the flask.
- Invert the flask.



(E) *Describe what happens.* _____

(F) *Explain why the water behaves as it does.* _____

_____

_____

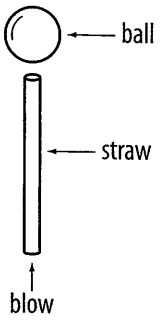
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31_BLOWING IT ALL AWAY-A

Daniel Bernoulli (1700–82) was a Swiss scientist who (among many other things) studied the relationship between moving air and air pressure. Bernoulli's principle states that the pressure in a fluid (i.e. a gas or liquid) decreases as its velocity increases. The following experiments illustrate this principle.

31.1 Up in the air

Materials: straw, table-tennis ball



Procedure

- Blow through a straw which you are holding vertically upward.
- Put a table-tennis ball in the stream of moving air above the straw.
- Blow strongly and keep the straw vertical.

(A) *Describe what happens.* _____

(B) *Complete the following passage. Use these words: high, pressure, fast, lower.*

A _____ moving stream of air has lower pressure, so the _____ above the straw while you are blowing is lower than in the normal air around it. Air moves from _____ pressure to low pressure, so the ball is constantly pushed into the area directly above the straw where the pressure is _____.

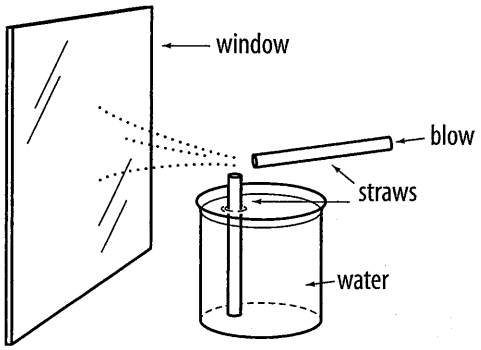
- Place the table-tennis ball on the bench.
- Use the straw to blow a stream of air to one side of the ball.

(C) *Describe what happens to the ball.* _____

(D) *Explain why the ball behaves as it does.* _____

31.2 Make a spray gun

Spray guns are used to deliver a fine mist of liquid, such as paint. This experiment shows the principle behind one type of spray gun.



Materials: beaker, two straws, window

Procedure

- Hold a straw in a beaker of water so it projects about 2 cm out of the water.
- Use another straw to blow at the top of the first straw as shown.

(E) *Describe what happens.* _____

(F) *Explain the principle behind your 'spray gun'.* _____

_____

(G) *Imagine you are going to paint something using a spray gun. Outline what advantages and disadvantages it would have over using a brush.* _____

31.3 Lifting the roof

When you are standing on the footpath by the side of the road and a large truck passes by quickly, you feel as if you are sucked towards the road. This is partly due to the Bernoulli Effect created by the moving streams of air. The tarpaulin over the back of a truck is flat when the truck is not moving, but it bulges upward when the truck is moving because of the air flowing over the top of it. Some houses have lost their tiles when there has been a strong wind. This has also happened to houses near airports when jumbo jets have flown overhead.

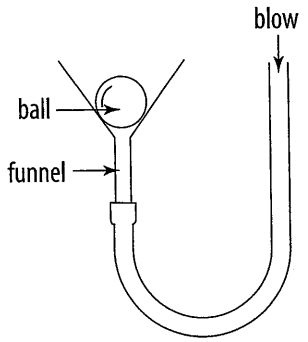
(H) *Explain how a strong wind can lift tiles from a house roof, mentioning the Bernoulli Effect in your answer.* _____

_____

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31.4 Huff and puff

Can you blow a table-tennis ball out of a filter funnel?



Materials: table-tennis ball, plastic funnel, tubing

Procedure

- Put a table-tennis ball into a plastic filter funnel.
- Attach a length of plastic tubing to the funnel.
- Hold the funnel vertically upright.
- Blow through the tubing.

(I) *Describe what happens.* _____

(J) *Explain why the ball behaves as it does.* _____

_____

(K) *Predict what would happen if you turned the funnel upside down while blowing through it.*

_____

- Now try this experiment with the funnel upside down.

(L) *Describe what happens.* _____

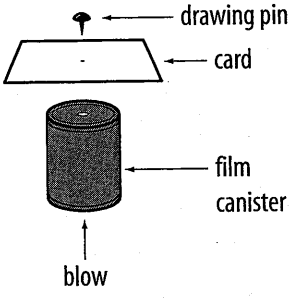
(M) *If it gets stuffy in a moving car, a good way to get rid of the stale air is to open a window slightly. Explain how this works.*

_____

_____

31.5 Up and down?

This experiment has many things in common with the previous one. Before you do it, forecast what will happen.



Materials: black plastic 35-millimetre film canister, drill, drawing-pin, playing card

Procedure

- Make a pencil-sized hole in the base of a film canister using a drill or similar device.
- Insert a drawing-pin through the centre of a playing card.
- Loosely position the point of the drawing-pin in the hole in the canister.
- Blow through the canister from the other end (as shown).

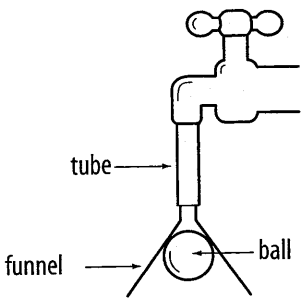
(N) *Describe and explain what happens.* _____

_____

31.6 Down and up?

The next experiment demonstrates the Bernoulli Effect in water.

Materials: table-tennis ball, tap, sink, tubing, filter funnel



Procedure

- Use a rubber tube to connect the funnel to a tap.
- Put a table-tennis ball in the* funnel.
- Hold the ball in place.
- Turn the water on.
- Let the table-tennis ball go.

(O) *Describe what the table-tennis ball does.* _____

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ANSWERS - A

1 Magnetism in three dimensions

(E) Paperclips are attracted to the nail.

(F) The paperclips fall off.

(G) A magnet that can be switched on and off, an electromagnet.

2 Invisible fields

(A) The compass needle will swing around.

(B) The compass needle would swing around in the opposite direction.

(C) The compass needle swings around in the opposite direction.

(D) The compass needle swings around in the opposite direction.

(E) The compass needle swings around in the opposite direction.

(F) magnetic, current, field, off

(G) A compass doesn't quite point to true north, so people need to make allowances when navigating.

(H) The line about which the Earth rotates.

(I) The picture will become distorted when a magnet is brought near it.

(J) Moving electrons create a magnetic field which is affected by the bar magnet.

3 Are you attracted?

(A) The paperclip will be suspended in midair.

(B) The paperclip will remain suspended.

(C) touch, attracted, magnetic, pass

(F) iron: Fe, cobalt: Co, nickel: Ni

4 More magnets

(A) Along a north–south direction.

(B) A compass

(C) It always points to the same end.

(D) They are attracted.

(E) A magnet

(F) The compass needles will point in opposite directions.

(G) The can has been magnetised by the Earth's magnetic field.

5 Magnetism on the move

(A) Both will have two distinct regions, but there will be more of a gap between poles with the horseshoe magnet.

(B) The iron filings will stand up and follow the magnet, leaving the sand behind.

(C) The magnet attracts iron filings but not sand. The magnetic field can pass through the card.

(D) It floats above the other magnet.

(E) Repulsion will cause it to balance the downward pull of gravity.

(F) Iron nail and steel pin

(G) No, all magnets must have a North Pole and a South Pole.

6 Electricity in motion

(B) If the globe lights up, it is a conductor, if it doesn't it is an insulator.

(C) All are metals.

(D) cables, insulators, glass, protect

(E) No

(F) It is not a complete circuit.

(G) It goes on and then off.

(H) The circuit was closed and then open.

(I) Switch globe on from more than one location.

(J) The buzzer will sound.

(K) The thick, central piece of cardboard separates the two sheets of aluminium foil. The circuit is not complete until the two sheets of foil are pressed together.

(L) Burglar alarm, shop customer alert.

(M) The strip will bend.

(N) The strip straightens up again.

(O) All metals expand, but not to the same extent. One metal expands more than the other, causing the strip to bend.

(P) The globe goes on and then off.

(Q) The bi-metal strip bends when heated to complete the circuit and the globe goes on. When it cools, the strip straightens and breaks the circuit.

(R) A fire alarm

7 Circuits with a difference

(A) For example, find out who is the quickest to move the ring from one end of the coat hanger wire to the other without buzzing.

(C) Small sprays of sparks will be produced.

(D) current, steel, sparks

(E) The steel wool will glow red hot and begin to disintegrate.

(F) The electricity causes the steel to heat up and then react with oxygen in the air.

8 Getting charged

(A) The comb will attract the cotton.

(B) The charged comb attracted the thread.

(C) Electrostatic attraction will make the ball roll towards the ruler.

(D) No, it wouldn't move, it would be too heavy.

(E) The stream of water will bend as it is attracted towards the charged balloon.

(F) The can will follow the balloon across the bench.

(G) False

(H) True

(I) The balloon will stay touching the ceiling.

(J) The balloon has become negatively charged when rubbed. Charged objects are attracted to uncharged objects, so the balloon is attracted to the ceiling. This force of attraction is strong enough to prevent the balloon from falling to the floor.

(K) They will repel each other.

(L) They have the same electrical charge.

(M) They repel each other.

9 The Van de Graaff Generator

(A) So that the charge is not carried away.

(B) It rises up into the air.

(C) It will strike at your hand.

(D) The paper eel becomes charged. As charged objects are attracted to uncharged objects, it moves towards your hand when it comes near.

(E) The confetti will rise into the air and scatter in all directions.

(F) Like charges repel. The pieces of confetti become charged with the same charge as the generator, so they are repelled by the generator and by each other.

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(G) The pie dishes will rise into the air one after another.

(H) The second dish will rise up and slip off the dome because it picks up the same charge as the first, and so on.

10 Organic electricity

(B) Zinc

(C) Copper, but lead close behind

(D) Same size of electrodes, same depth of insertion, etc.

(E) Bubbles of (hydrogen) gas

(F) After a minute or so a blue colour will appear at the wire connected to the positive terminal.

11 I don't want to change

(A) The bottle will move with the paper.

(B) The bottle will stay where it is.

(C) The inertia of the bottle resists any change. The more quickly you pull the paper, the greater the inertia. When you pull the paper very quickly, the inertia is sufficient to overcome the friction between the paper and the bottle.

(D) It would move with the paper because it has less mass, so less inertia.

(E) Water moves with glass.

(F) Water stays put.

(G) First time, cord will break between mass and point of attachment. Second time, cord will break between mass and pull.

12 Bumping into each other

(A) The basketball will not bounce much, but the tennis ball will shoot into the air.

(B) The momentum from the large ball is transferred to the small ball. The small ball has less mass so, in order to conserve momentum, its velocity increases.

(C) No, they are too small so their momentum is too small.

(D) The marble at the other end will roll away.

(E) Two marbles will move from the other end.

(F) One coin from opposite end will move.

(G) Two coins from opposite end will move.

(H) One ball at the opposite end will move out and back, then the original ball will move out, and so on, but with decreasing movements until all the balls are still.

(I) Two balls at the opposite end will move out and back, then the original two balls will move out and so on, but with decreasing movements until all the balls are still.

(J) Kinetic energy is changed to other energy, such as sound and heat.

(K) Whichever ball bounces the highest

(L) Plasticine

(M) Deformed

(N) Crumple zones slow deceleration and use up kinetic energy, so they make cars safer for passengers.

13 Rocket science

(A) Water pours out of the holes and the carton starts to spin around.

(B) The water going out one way causes an equal and opposite force pushing the carton the other way.

(C) The force of the water coming out of the nozzle pushes the hose in the opposite direction.

(D) Water comes out of the neck of the bottle. It should travel at least 10 metres.

(F) The change in momentum of the water equals the change in momentum of the bottle. The bottle has less mass than the water, so it moves in the opposite direction at a relatively greater velocity.

(G) To slow them down so they don't sustain damage on landing.

(I) After a few moments, the canister will shoot upwards with a 'pop'.

(J) Carbon dioxide gas builds up until the gas pressure exceeds the frictional force between the canister and its lid. Momentum is conserved during the take-off; as the canister has much less mass than the lid/earth system, it moves into the air at a higher velocity.

(K) Other fish move by moving their fins and also swivelling their bodies.

(L) Potato will shoot out of the syringe.

(M) Pressure builds up and the force finally exceeds the force of friction between potato and syringe.

(N) For example, it is done that way for dramatic effect, or perhaps the shooter is expecting the recoil and is braced for it.

(P) Simpler, cheaper, quieter, don't alarm other prey.

(Q) Both would move away from each other in opposite directions at the same speed.

14 Swinging

(A) Mass C should start to swing.

(B) Mass D should start to swing.

(C) C should start A swinging, D should set B swinging and E should show little effect.

(D) Mention resonance, causing similar lengths to move in response to each other.

(E) Various curves depending on the length of the cord.

15 All in a spin

(A) You will feel the wheel resisting the change in direction.

(B) The spinning wheel has inertia that maintains the direction in which it is aligned. To change the direction of the spinning wheel you need to overcome this inertia.

(C) The wheel resists movement.

(D) The bigger the wheel, the greater the inertia.

(E) The LP resists the swinging movement.

(F) Similar answer to the spinning bicycle wheel.

(G) The drawing pin spins for a while.

(H) Its inertia will make it remain upright as long as it is spinning strongly.

(I) The speed will increase as you draw your legs in closer.

(J) conserved, decreases, angular

(K) The speed will increase when you move to the centre.

(L) Ice skating

(M) The padlock will fall towards the floor but will not hit it. The key swings and wraps the string around the pencil.

(N) The length of its string is continually decreasing because the padlock is pulling it, so the key keeps speeding up. Its velocity increases so much that it wraps the string around the pencil.

(O) The key doesn't wrap round the pencil in either case.

(P) You have moved too.

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ANSWERS - C

16 Centripetal force

(A) A force pulling outwards.

(B) Centripetal force

(C) It would move away at a tangent from the radius of the circle of rotation.

(D) The brick will be lifted off the ground.

(H) True

(I) Mention circular motion, inertia and centripetal force.

(J) The nut will continue to circle around the balloon. There may be a 'howling' noise, depending on the size of the nut and the size and tension of the balloon.

(K) The coin will circle more smoothly than the nut.

(M) For example, an outward force exerted by a body moving on a circular path on the object holding it.

17 Bumping into things

(B) It should be a paper tube. The one where the bolt moves least. Observations should note that paper has crumpled, etc.

18 Bridging the gap

(A) Rivers, valleys, lakes, roads, railways

(B) Steel

(C) Natural occurrences that include weathering, earthquakes, strong winds, and extremes of temperature.

(D) Results will vary, but the straight piece of card will be the weakest.

(E) Mention arches with cables holding roadway.

19 A matter of gravity

(A) If air resistance is the same, they should hit the ground at the same time, regardless of weight. Air resistance and wind will have a noticeable effect on very light balls, such as table-tennis balls.

(B) The water will not come out of the hole while the cup is falling.

(E) They will both hit the ground at the same time.

(F) Gravity affects them the same.

20 Doing things mechanically

(B) Very similar

(C) It changes the direction of a force without changing its value.

(E) Force should move twice as far as load, force should be half (approximately) the load.

(F) Force should move four times as far as effort, force should be quarter (approximately) of the load.

(G) The two broomsticks will be pulled together.

(H) Because of the friction between the rope and broomsticks.

(I) Simpler equipment, but less efficient

21 A friction story

(A) Sandpaper

(B) It generated the most heat.

(C) It was the roughest.

(E) The friction between the pages will make them very hard to pull apart.

(G) Mass of shoe, type of sole.

22 Mobile machines

(A) Movement

(B) Stored energy

(C) Elastic band unwinds, paddle wheel turns and 'boat' moves forward.

(D) Kinetic/movement

(E) Potential/elastic

(F) Movement/kinetic

(G) Yes, because it changes one form of energy to another.

(H) As the trap closes, the potential energy in the spring is converted to kinetic energy. This movement pulls the string, which is wound around the axle, so the car will move forward.

(I) The can will roll to a stop about one metre away and then roll back to you.

(J) As the can rolls away from you, the lead weight does not roll (due to inertia), so the string winds up. In this way kinetic energy is changed to potential energy, which is stored as tension in the string. When the can rolls back, the potential energy is converted back to kinetic energy.

23 Feeling the pressure?

(A) The card and water will not fall.

(B) The external air pressure pushing on the card is greater than the downwards water pressure on the card.

(C) You will initially suck up a small amount of fluid and then no more.

(D) No air was able to get into the bottle to equalise the air pressure.

(E) The water will not pour out.

(F) The water will fall from the hole.

(G) Initially, the external air pressure holds the water in. Only when the top is open can the water escape, due to air being able to replace the water pouring out.

(H) There would be less wind because of the holes in the fly swatter, so the fly might not notice it in time.

(I) For example, millibars, millimetres of mercury, kilopascals, hectopascals, pounds per square inch (psi).

(J) True

24 Keeping up the pressure

(A) The water stays inside the test tube.

(C) The ruler will break.

(D) Force at exposed end where it is hit, plus weight of paper plus air pressure.

(E) To maximise the air pressure.

(F) The desk is lifted.

(G) A partial vacuum will form between the plunger and the desk. The outside air pressure will be strong enough to keep the plunger attached to the desk as it lifts.

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25 Relieving the pressure

(A) The can will collapse with a loud bang.

(B) When the water is heated, it turns to steam and pushes air out of the can. When the can is cooled suddenly, the steam condenses. This causes a drop in pressure inside the can, and the outside air pressure squashes the can.

(C) Teacher to check

(D) Teacher to check

(E) Teacher to check

(F) The bottle will be partially crushed.

(G) The hot water causes the air to expand and some is pushed out of the bottle. When cooled, the air contracts causing the pressure inside the bottle to fall. The outside air pressure is greater than inside air pressure, which causes the bottle to decrease in volume.

(H) The balloon will be pulled inside out into the flask.

(I) The air inside the flask cools and contracts. External air pressure forces the balloon into the flask.

26 Changes in air pressure

(A) High pressure means stable weather. Low pressure means unstable weather.

(B) Teacher to check. Reasons could be: more local, more up to date, greater resources.

(C) Flames go out, the egg goes into the flask.

(D) The burning paper heats the air in the flask, so some of the air is forced out by expansion. As the air cools it contracts, and the greater ambient air pressure will force the egg into the flask. When the oxygen is used up or the paper is totally burned, the flames will go out.

(E) The marshmallow will inflate. As the bell jar is evacuated, the air in pockets inside the marshmallow has a higher pressure than the air in the bell jar. The marshmallow gets bigger as the air inside it expands.

27 It's a gas!

(B) Will depend on concentration, but usually bicarbonate reacts more quickly.

(C) Bicarbonate should be faster.

(D) Exhaled breath, indicator changes more quickly with breath than with the atmosphere.

(E) Concentrations of indicators, delivery of same volume of CO_2 to each indicator.

(F) Carbon dioxide

(G) Calcium hydroxide

28 Putting pressure on air and water

(B) Stays the same

(C) The pen top will sink when you squeeze the bottle and rise again when you stop squeezing.

(E) squeeze, air, compressed, sinks

(F) Press rubber top to send diver down, release pressure for it to come up.

(G) True

(H) True

(I) Yes, a gas coming out of a solution when pressure is lessened.

(J) Go all the way around.

29 Fluids under pressure

(A) Moves outwards.

(B) Should be the same.

(C) Smaller one should move further.

(D) Your finger will be unable to resist the movement.

(E) The bricks will fall over.

(F) The weight of the water in the plastic tube exerts pressure on the water in the hot-water bottle. This pressure is multiplied over the whole inner surface of the hot water bottle and is enough to lift the bricks.

30 Water falls

(A) The water will drain from the upper beaker to the lower one.

(B) Teacher to check. Should show arrows pushing down on top beaker.

(C) Danger from fumes, poisonous if swallowed.

(E) As water leaves the flask to the sink, more water sprays into the flask through the thin nozzle.

Could also use coloured water in the flask to show the flow more clearly.

(F) As the water in the flask drains into the sink, a partial vacuum is created in the flask. Ambient air pressure pushes the water from the beaker into the flask.

31 Blowing it all away

(A) The table-tennis ball will remain suspended just above the straw as long as air is coming out strongly.

(B) fast, pressure, high, lower

(C) The ball will initially move towards the stream of air.

(D) Moves to area of low pressure.

(E) A fine spray of water will hit the glass.

(F) The fast-moving stream of air creates lower pressure above the vertical straw. The higher ambient air pressure acts on the water in the beaker, forcing some of it up the straw.

(G) Smoother finish, can cover large areas quickly, need to wear mask, more expensive than brush.

(H) The rapidly moving air causes the pressure above the tiles to be less than under the tiles. The difference in pressure causes the tiles to lift.

(I) The ball will remain in the funnel.

(J) The air you are blowing is flowing between the ball and the sides of the funnel. This fast-moving air has lower pressure than the air outside the funnel, so the ball is constantly pushed into this area where the pressure is lower.

(L) The ball will not fall out of the funnel as long as you blow strongly.

(M) There is a stream of air flowing down the outside of the car, so the air pressure is relatively lower outside and the air is forced out of the window.

(N) There is a low pressure area between the card and the canister because the air is moving quickly there. The higher air pressure on the other side of the card will hold it in place.

(O) The ball will remain in the funnel.

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TEACHER'S NOTES

1.1 As the size of the container is variable, so too will be the amount of iron filings to be used. As a rule of thumb, a level teaspoon is enough for a 500 mL container.

1.2 If using a power supply, make sure that the current isn't switched on for too long.

2.1 Don't leave the power supply unit on for more than a few seconds.

4.4 This might best be done as a demonstration by the teacher if the number of cans needed is a problem.

5.1 Iron filings will cling to the magnet, so it's a good idea to put magnets in plastic bags.

6.1 Solid materials for testing could include metal strips, glass, plastic, wood, paper. You may need to check the actual voltage required – it will depend on the particular globe used.

6.5 Complete the circuit before heating to ensure the globe is working. You could also replace the globe with a buzzer.

8.7 Perspex is clear, ebonite is black. For charging the rods, you can use rayon or any other smooth, shiny synthetics.

10.1 Readings are variable, you may need to use a milliammeter.

13.1 You may need to explain the exact position of holes to be punched. The class can also investigate what happens if different combinations of holes are used.

14.1 Do not swing the masses in line with the suspending cord, but across (perpendicular to).

14.2 There are many variables involved. The relative sizes of cord may have to be determined experimentally to get the 'best' result. Take care with possible mess made by the sand.

16.3 There are many alternative ways to do this experiment. This is one that can be done inside.

17.1 Almost anything that rolls can be used for the 'car'.

18.2 You may have to vary the distance between books.

20.1 You need to ensure that the spring balance/force meter is calibrated in grams.

20.2 This is a tricky procedure. You should allow a fair amount of time to do this. You will need to ensure that there is tension on the cord at all times. This is best done with at least two students holding and helping.

21.1 You can also use a range of other surfaces, such as wet glass. Make sure students don't rub too hard on the sandpaper.

22.2 You could use a mouse trap and show (with great caution) how it works.

24.3 This works best if the desk has a smooth, laminated top.

25.3 Students may need a bit of practice to get this to work.

26.1 Digital weather stations are cheap and also measure humidity. Some have a memory and keep a history over some hours. Using a data logger and probes/sensors would be an alternative approach.

The Australian Bureau of Meteorology (www.bom.gov.au) and other websites are good sources of radar images and other meteorological information.

26.2 To recover the egg from the flask, firstly turn the flask upside down so the egg is fitting snugly into the neck. Then put your lips around the mouth of the flask, to seal it tightly, and blow as hard as you can into it. When you stop blowing, the egg will move out of the flask and into your mouth, as you have increased the pressure inside the flask. Discard the egg after use.

27.1 Bicarbonate is not a specific test for carbon dioxide; it will react in a similar manner to any acidic gas.

29.3 An empty four-litre bladder from a wine cask can be substituted for a hot water bottle (with a bit of work).

30.3 Florence flasks are often referred to as boiling flasks, round-bottomed flasks or flat-bottomed flasks, depending on the individual shape.

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